



Review article

AI techniques for IoT-based DDoS attack detection: Taxonomies, comprehensive review and research challenges

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ABSTRACT

Distributed Denial of Service (DDoS) attacks in IoT networks are one of the most devastating and challenging cyber-attacks. The number of IoT users is growing exponentially due to the increase in IoT devices over the past years. Consequently, DDoS attack has become the most prominent attack as vulnerable IoT devices are becoming victims of it. In the literature, numerous techniques have been proposed to detect IoT-based DDoS attacks. However, techniques based on Artificial Intelligence (AI) have proven to be effective in the detection of cyber-attacks in comparison to other alternative techniques. This paper presents a systematic literature review of AI-based tools and techniques used for analysis, classification, and detection of the most threatening, prominent, and dreadful IoT-based DDoS attacks between the years 2019 to 2023. A comparative study of real datasets having IoT traffic features has also been illustrated. The findings of this systematic review provide useful insights into the existing research landscape for designing AI-based models to detect IoT-based DDoS attacks specifically. Additionally, the study sheds light on IoT botnet lifecycle, various botnet families, the taxonomy of IoT-based DDoS attacks, prominent tools used to launch DDoS attack, publicly available IoT datasets, the taxonomy of AI techniques, popular software available for ML/DL modeling, a list of numerous research challenges and future directions that may aid in the development of novel and reliable methods for identifying and categorizing IoT-based DDoS attacks.

1. Introduction

Advanced technologies of the future are progressively enhancing the digital landscape. Internet of Things (IoT) technology is one of the breakthrough technologies that is rapidly expanding due to the growing network of interconnected online devices. Internet of Things stands out as one of the most resilient technologies in today's digital era. The number of IoT devices is escalating every year as shown in Fig. 1. As per Fortune Business Insights, the value of the global IoT market was USD 544.38 billion in 2022 which will be expected to grow by USD 3352.97 billion by 2030. The forecasted compound annual growth rate (CAGR) is 26.1% from 2023 to 2030 [1].

IoT devices are used in different areas like smart homes, education, health care sectors, agriculture, smart vehicles, smart grid, etc. [2]. Homes and urban areas are becoming more intelligent to meet fundamental human requirements, including enhancing security, managing waste, improving air quality, and providing entertainment [3]. Manufacturers are generating several internet-connected IoT devices having inherent security vulnerabilities and flaws. The expeditious spread of

vulnerable IoT devices and the accessibility for attackers to discover them results in a continuous mushroom growth of cyberattacks like DDoS, phishing, spamming, click fraud, etc. Among these cyberattacks, DDoS is one of the most prominent attacks originating with the help of botnets made from vulnerable IoT devices.

In the first quarter of the year 2023, there was a sustained upswing in IoT-based DDoS attacks, with an increasing trend from the preceding quarter. These DDoS attacks have seen a 68% year-over-year growth over the time. Concurrently, experts at StormWall observed a distinct shift towards higher utilization of botnets. This was evident in the increased activity of well-known botnet families Mirai, Morii, Moris, etc. This trend aligns closely with the expanding proliferation of "Internet of Things", providing greater opportunities for botnets. Cybercriminals focused predominantly on finance (with 26% attack share), telecom (having 17% share), and entertainment (making 14% share) verticals to cause disruption and inflict harm on the core national economic pillars. Fig. 2 also shows the percentage increase in IoT-based DDoS attacks in the second quarter of 2023 as compared to 2022 and also mentions

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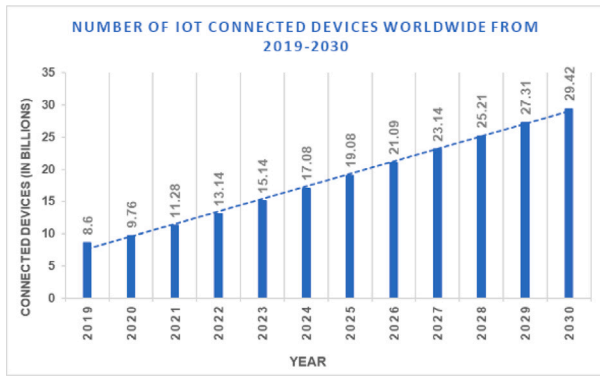


Fig. 1. Growth of IoT devices from 2019 to 2030.

the share of attacks by different industries like finance, Government services, health care, etc. The primary cause for the significant increase in these attacks is the spread of IoT devices, which are weaponized to form botnets [4].

IoT-based DDoS attacks are the most prevalent form of network disruption attack and one of the biggest challenges that security experts and IT consultants are facing. Some of the recent incidents of such attacks include:

- (1) As per Nokia's Threat Intelligence Report – 2023, DDoS traffic originating from insecure IoT Devices has grown 5-fold in 2023 over the past year to disrupt Network services. The number of IoT Devices involved in DDoS attacks increased from around 2,00,000 a year ago to approximately. 1 million devices which are generating more than 40% of all DDoS traffic today [5].
- (2) An ACK flood DDoS attack, which started from a Mirai-variant botnet comprised of thousands of IoT devices, was one of the biggest attacks in the second quarter of 2023. The target of the attack was an American Internet Service Provider. It peaked at 1.4 terabits per second (Tbps) [6].
- (3) Fodcha DDoS Botnet (2022): The Fodcha botnet is focusing on IoT devices for executing DDoS attacks. It is aimed at systems like DVRs, routers, and servers running various CPU architectures like MIPS, mpsl, arm, x86, and others. As per Netlab's findings, on October 11, 2022, the Fodcha Botnet reached a new peak, targeting 1396 entities within a single day. Using the Fodcha botnet, a DDoS attack affected a healthcare organization on June 7 and 8, 2022, and communication infrastructure on September 2022. The majority of Fodcha's targets are situated in China and the United States, but the botnet's impact extends worldwide, with compromised systems found in Europe, Australia, Japan, Russia, Brazil, and Canada [7].
- (4) On September 5, 2021, Yandex, the most popular Russian internet search engine, experienced a severe DDoS assault from a novel IoT botnet named Meris, which reached its peak at 21.8 million HTTP requests per second (RPS). The malicious traffic was generated by over 250,000 compromised IoT devices, predominantly consisting of network devices from Mikrotik [8].

The major contribution of this review paper is

- To study the targets and motivation of IoT-based DDoS attacks.
- The Layered architecture of IoT and various IoT security challenges are thoroughly discussed.
- The life cycle of botnets using compromised and vulnerable IoT devices and various existing botnet families are illustrated.
- The launching methodology of DDoS attacks originated from IoT devices, the taxonomy of various DDoS attacks, and attacking tools are discussed thoroughly.

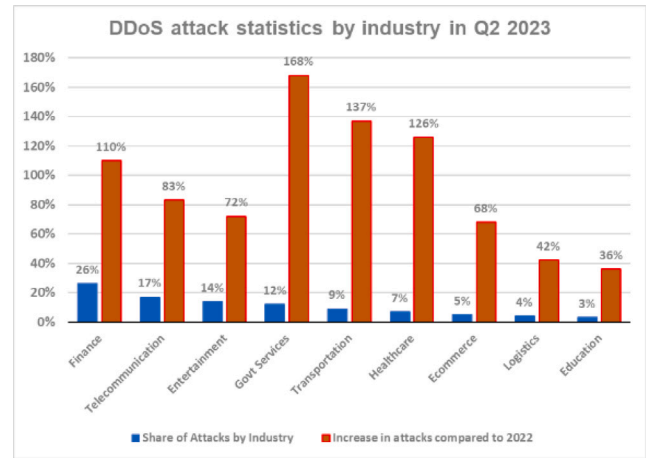


Fig. 2. DDoS attack in IoT networks statistics by different industries in Q2 2023.

- Comparative analysis of the existing literature employing various existing state-of-the-art Machine learning(ML) and Deep Learning(DL) classifiers to detect IoT-based DDoS attacks.
- Various popular software tools, frameworks, and libraries used to implement various ML and DL classifiers are thoroughly presented.
- Comparative analysis of various publicly available datasets with IoT Traffic has been discussed.

Various review papers have been published between the duration of 2018 to 2023 that describe DDoS attacks in IoT networks. But many aspects are missing in some of them as shown in Table 1. All the survey papers are thoroughly investigated. Most of the papers have covered a few perspectives of IoT in DDoS attack detection. Many perspectives like IoT botnet formation, various botnet variants, DDoS attack tools, publicly available IoT datasets, popular software tools and frameworks for machine learning, and deep learning classifiers specifically for DDoS attack detection are missing in the previous surveys and review papers. All these perspectives are thoroughly studied and explained in this systematic literature review paper. Moreover, the study has exclusively compared and investigated existing related studies using state-of-the-art machine learning (ML) and deep learning (DL) techniques to detect IoT-based DDoS attacks.

2. Targets and motivation

Recent market research indicates that over the past year, IoT based DDoS attacks have been intensified both in frequency and volume. The trend shows shorter attack duration with larger attack vectors. DDoS attacks can impact individuals, businesses, banks, Internet service providers, and even government entities. Financial gain is the primary motivation, but political administrations and governing authorities are also frequent victims. The motives and psychological factors causing DDoS attacks are diverse, encompassing financial gain, seeking revenge, ideological convictions, engaging in cyber warfare, or even finding personal amusement as their basis which are listed as follows:

- Financial and Economic Gain
- Intellectual Challenge
- Retaliatory action
- Ideological belief
- Cyberwar
- Personal Amusement
- Protest and Opposition

Table 1

Comparison of existing review studies (2018–2023) (see [9–15]).

Author	Pub.Year	Security issues of IoT layers	Botnet life cycle	Various botnet families	IoT-Based DDoS attack taxonomy	DDoS attack tools	Comparative study of ML/DL of DDoS detection	Study of IoT dataset	Study of S/W Framework/Tools for ML/DL classifiers
De Donno et al. [9]	2018	×	×	✓	✓	×	×	×	×
Lohachab and Karambir [16]	2018	✓	×	×	✓	×	×	×	×
Roohi et al. [10]	2019	✓	×	×	✓	×	×	×	×
Vishwakarma and Jain [17]	2020	✓	×	×	✓	×	✓	×	×
Irum et al. [11]	2020	✓	×	✓	✓	×	×	×	×
Al-Hadhrani et al. [12]	2021	✓	×	×	✓	×	✓	×	×
Ashraf et al. [13]	2021	×	×	✓	✓	×	✓	×	×
Varalakshmi et al. [14]	2022	×	×	×	✓	×	✓	×	×
Kumari and Jain [18]	2022	✓	✓	✓	✓	×	✓	×	×
Alahmadi et al. [19]	2023	×	×	×	✓	×	✓	×	×
Shukla et al. [15]	2023	✓	×	×	✓	×	✓	✓	×
Proposed study	2023	✓	✓	✓	✓	✓	✓	✓	✓

3. Systematic literature review process

SLR is defined as a “systematic, explicit, and reproducible method for identifying, evaluating, and synthesizing the existing body of completed and recorded work made by researchers, scholars, and practitioners” [20]. The procedure follows in the present study is as per the guidelines of [21]. It focuses on IoT layered architecture, security challenges of IoT Networks and IoT-based DDoS attack detection using AI techniques from 2019 to 2023. A comparative analysis of IoT traffic benchmark datasets has been done in this SLR. Various knowledge gaps and promising future directions are identified from this systematic literature review that will be helpful for fellow researchers. Various steps followed in this SLR are explained as follows:

3.1. Objectives and research questions

The primary objective of this paper is to investigate the emerging threat of IoT-based DDoS attacks and comparative analysis of AI-based detection techniques being utilized by the researchers. Therefore, the initial step includes gaining insight into IoT networks, layered architecture, security challenges, and recent large-scale attacks that have interrupted them. Subsequently, we have thoroughly investigated various advanced AI techniques employed by researchers to analyze network traffic, distinguishing between benign and malicious IoT-based DDoS traffic patterns. To carry out a systematic literature review, various research questions have been framed as mentioned below.

RQ1: What is the motivation behind considering IoT-based DDoS attacks?

RQ2: What is the layered architecture of IoT networks and an array of security challenges heightening the vulnerability of IoT devices to potential DDoS attacks?

RQ3: What is the Life cycle of IoT Botnet and various botnet variants causing the IoT devices as malicious?

RQ4: What is the attack launching mechanism, attack tools, and Taxonomy of IoT-based DDoS attacks?

RQ5: What are the existing state-of-the-art IoT-based DDoS attack detection AI techniques and their comparative analysis based on various parameters?

RQ6: What are the existing machine learning (ML) and deep learning (DL) software frameworks and libraries to design model for IoT-based DDoS attack detection and their comparative analysis?

RQ7: What are the various publicly available IoT traffic datasets and comparisons between IoT traffic and Non-IoT traffic datasets?

RQ8: What are the various research gaps in the existing studies?

Section 2 represents the motivation behind considering IoT-based DDoS attacks, Section 4 states the layered architecture of IoT networks, Section 5 includes major IoT security challenges, Section 6 explains the Life cycle of IoT Botnet and various botnet variants, Section 11 describes the existing state-of-the-art IoT-based DDoS attack detection AI techniques during 2019 to 2023, Section 12 states various publicly available IoT traffic datasets and different research gaps are analyzed in Section 14.

3.2. Search criteria

The following specified search criteria have been used to answer the above-mentioned research questions with the help of pertinent literature:

- (1) Create a keyword list based on the research questions.
- (2) Find keywords in selected data sources.
- (3) Distinguish various keyword variants and spellings.
- (4) Use the Boolean operators “AND” and “OR” to correlate essential words and concepts.

The Search string used to search the relevant literature is (IoT OR “internet of things”) AND (DDoS OR “distributed denial of service”) AND (“machine learning” or “deep learning”). All searches are based on the abstract, title, and keywords of the publications to prevent obtaining a large number of unnecessary search results. The aforementioned search string was employed in all the chosen data sources for the final selection process of primary studies to narrow down the search criteria.

3.3. Data sources

In this study, we have used the subsequent data sources to extract pivotal research articles from the existing literature. These data sources are:

- (1) Science Direct (<http://sciencedirect.com/>)
- (2) Springer (<http://link.springer.com/>)
- (3) IEEE Xplore (<http://ieeexplore.ieee.org/>)
- (4) Google Scholar (<http://scholar.google.com>)

3.4. Inclusion and exclusion criteria

We framed the search criteria with the subsequent points to be included in order to search specifically the current trends of defense against IoT-based DDoS attacks, :

IC1: The studies published between 2019 to 2023.

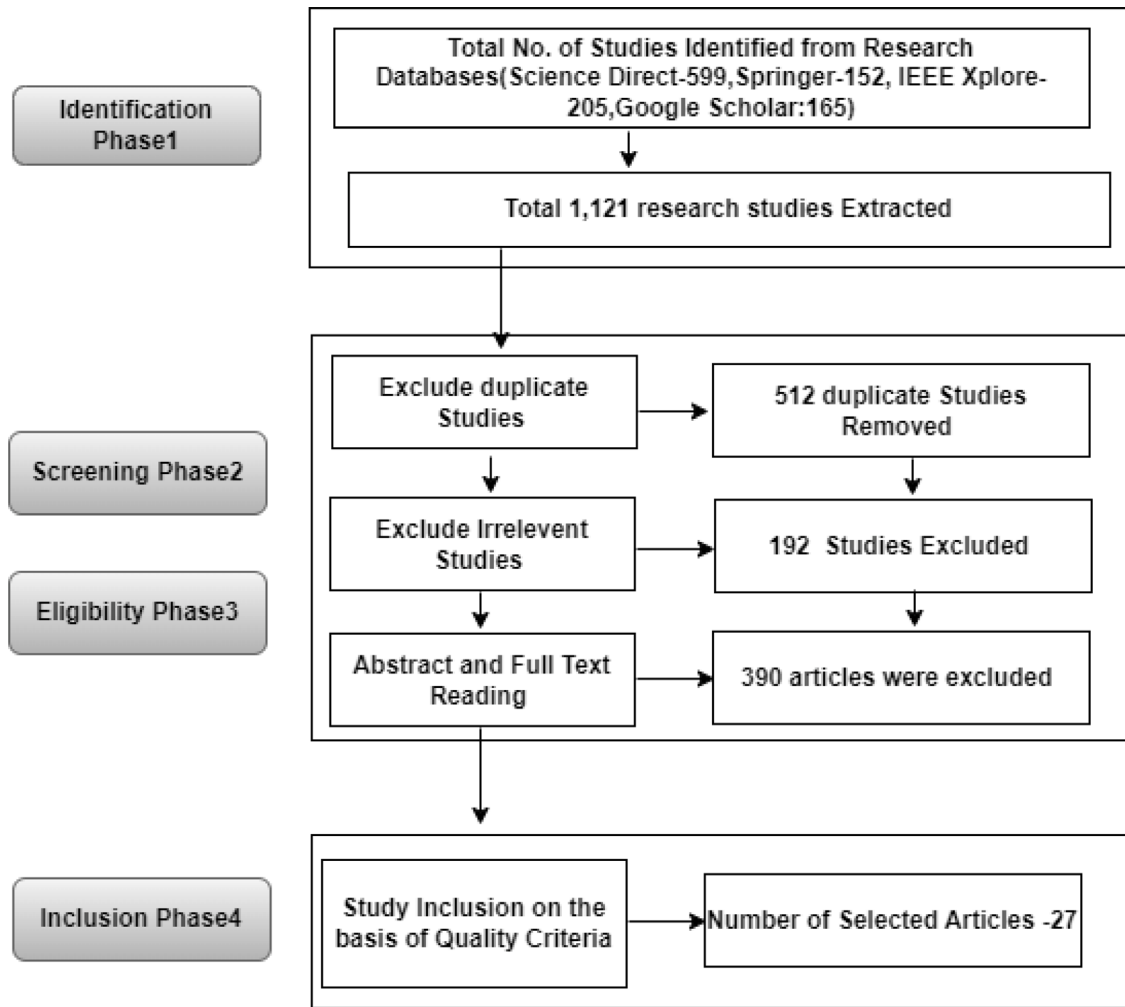


Fig. 3. Systematic literature review process.

IC2: The article published in Journals with SCI/Scopus indexed or conferences or book chapters.

IC3: Include studies published in the English language only.

IC4: Include studies that discuss techniques, strategies, and models for identifying IoT-based DDoS attacks.

IC5: Articles that comply with the research questions.

The exclusion criteria applied in this review are as follows:

EC1: Exclude study that is not available in printed or electronic format.

EC2: Exclude studies that are published in Thesis, Workshops, etc.

EC3: Non-conference and non-journal research studies are excluded.

EC4: Research studies that followed some already existing findings with little or no modification.

EC5: Duplicate research studies.

Above mentioned inclusion and exclusion criteria helped us to remain specific during the survey and thus to come up with relevant data to complete the study.

3.5. Quality assessment criteria

The selected papers were screened based on the quality criteria mentioned in Table 2:

3.6. Selection procedure

The selected search string is applied to the research data sources Science Direct, IEEE, Springer, and Google Scholar. The flow chart as

Table 2
Quality assessment criteria.

Quality assessment	Questions
QA1	Is the research goal indicated in the study?
QA2	Is the research well-cited?
QA3	Is the research title relevant to the research questions?
QA4	Is the study answering of all the research questions?

shown in Fig. 3 describes the Systematic literature Review process of this review study. The flow chart is divided into four phases. These phases elaborate the whole selection procedure of research articles. In the identification Phase, 1121 research articles were extracted from the selected data sources. In the Screening phase, 512 duplicate studies were excluded from the total extracted studies. In the Eligibility phase, 192 articles were not relevant as per the defined research questions, and these articles not fulfilling the above-mentioned inclusion and exclusion criteria. Subsequently, 390 research studies are excluded after the abstract and full-text reading. Finally, 27 studies were selected that fulfilled the inclusion criteria in the last Inclusion phase.

4. Layered architecture of IoT

Various functions are executed across different levels of IoT operations to accomplish the specific goals of smart applications. Different

reference models and frameworks have been introduced in the literature to gain a thorough understanding of the intricate workings of IoT. The fundamental layered architecture with five layers [22] is shown in Fig. 4.

4.1. Perception layer

The perception layer focuses on data acquisition through various sensing devices such as sensors, actuators, RFID readers, edge devices, and smart controllers. The data retrieved in this stage must adhere to standardized formats to ensure compatibility with different protocols employed within the network. This layer is also referred to as the edge device layer.

4.2. Adaptation layer

An adaptation layer is designed to add protocols for low-power IoT devices to efficiently use IPv6 addressing. This layer utilizes the 6LoWPAN protocol, enabling closed, energy-efficient networks to intelligently connect to larger public networks like the Internet with minimal computational and battery requirements.

4.3. Network layer

The next layer is dedicated to networking and controls the communication between the edge devices and the application responsible for data sensing. This layer exhibits various wireless mediums like Bluetooth, Wi-Fi, 3G, Zigbee, and others to transmit the gathered data. Data is packaged and sent to the next layer for further processing.

4.4. Transport layer

The transport layer guarantees reliable end-to-end communication and manages data segmentation, reassembly, and detection of errors. The protocols used in this layer are TCP and UDP. It provides an interface between networks like 3G, Zigbee networks, and radio-frequency identification (RFID) and allows transmission between them. This layer also ensures the security of IoT networks.

4.5. Application layer

The Application layer constitutes the outermost and concluding tier. Various smart applications leverage IoT as a conduit, encompassing domains such as smart cities, connected cars, smart health, smart homes, smart grids, and more on this layer. This layer interacts with the end-users. It provides access to the data which is processed in the other layers.

While its definition appears simple, the intricacy and significance of IoT expand significantly when addressing its security and privacy concerns. Factors contributing to its sensitive vulnerability include the utilization of unreliable networking protocols, reduced human intervention, limitations in power resources, and memory constraints. These factors render IoT systems susceptible to a wide array of security threats, including sophisticated attacks prevalent in the contemporary technological landscape [17].

5. IoT security challenges

Despite the potential for improving the current network paradigm, there are certain IoT security issues and challenges that require thorough investigation [23]. The Internet of Things (IoT) offers efficient and innovative services to its end users, but it meets significant security and confidentiality challenges arising from the weaknesses of devices and the heterogeneity of networks. Different IoT security and privacy challenges are shown in Fig. 5.

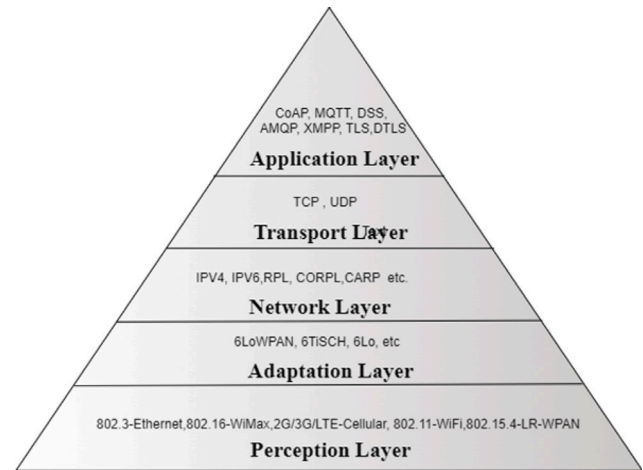


Fig. 4. IoT architecture and protocols.



Fig. 5. IoT security challenges.

5.1. Authentication

IoT devices have weak authentication. They have diverse resource constraints like low power, and low battery and do not have adequate storage to implement various cryptographic algorithms to handle authentication making them vulnerable to attacks. It will be helpful for the adversary to exploit the IoT devices and get into the network [24].

5.2. Privacy

Internet of Things (IoT) devices exhibit unreliable interfaces, deficient authentication mechanisms, inadequate encryption, and vulnerable services of networks. These vulnerabilities jointly enable adversaries to gain unauthorized access to safeguarded data that may have been collected without legitimate necessity [17].

5.3. Trust

The IoT ecosystem operates within a diverse network comprising numerous devices and actuators used for data collection. This diversity

gives rise to trust concerns among both the users and devices, and trust is dominant in ensuring the security and dependability of IoT services. An IoT network must maintain principles of confidentiality, integrity, and availability (CIA Triad) to design a robust model [25].

5.4. Access control

Access control plays the vital function of authorizing only legitimate users to gain access to the devices, information, and network resources within the network environment. However, the Internet of Things utilizes low-powered and unreliable network devices characterized by constrained resources, including power and bandwidth. Therefore, implementing effective access control becomes a significant challenge within the IoT environment [18].

5.5. Confidentiality

The mushroom growth of IoT devices has resulted in a rapid increase in data generation and a substantial volume of information. However, the intrinsic limitations of these devices introduce substantial challenges when managing a huge amount of data at the perceptual level. Consequently, the data collection layer transfers this data to the cloud, where preserving the confidentiality of this information becomes a complex endeavor [26].

5.6. Device security

A variety of devices are interconnected in the IoT networks with different platforms and architectures that communicate over various networks. Regardless of the security measures in place, the system inherently offers limited control, as an attack on any one of these devices can potentially compromise the overall security [18,27].

5.7. Identity management

IoT networks comprise a multitude of interconnected devices, and managing them presents a formidable task. The challenge lies in dynamically allocating distinct identifiers to all these devices and maintaining them. The IPV6 architecture can be employed to facilitate identity management, which offers auto-configuration features [18].

5.8. Integrity

Data may undergo modifications and alterations during transmission to the recipient, leading to an unreliable service within the IoT system. In numerous scenarios and applications, it is imperative to ensure data integrity between IoT devices. The tampering of data and information while in transit can have significant adverse consequences as very sensitive data remains under the control of IoT devices.

5.9. Availability

Irrespective of where IoT devices are situated or when they require resources, these resources must remain accessible to physical entities. In order to achieve worldwide utilization of IoT platforms, resources, and services, a multitude of small IoT networks are interconnected with them in a distributed fashion. Consequently, ensuring accessibility becomes a critical concern [28].

5.10. Non-repudiation

Non-repudiation is a crucial security facet in the IoT ecosystem that ensures the sender and receiver can transfer the information without tampering. It also guarantees the indisputable transfer of data or identities between two IoT devices. Non-repudiation assures the source node that its data transmission is secure and the receiving node that the received data matches as sent by its source [29].

5.11. Scalability

Scalability denotes the capability of the IoT network to incorporate extra services, tools, and devices into a system as it evolves without compromising its performance. In the context of IoT, a major challenge is accommodating a vast array of devices with diverse memory, computing power, storage, and bandwidth capacities.

5.12. Adaptability

Security in IoT networks is challenged by the need for adaptability. As IoT technologies evolve and new devices are introduced, it becomes essential to ensure that security measures can effectively adjust to these changes. Neglecting adaptability can result in vulnerabilities and potential security breaches.

6. IoT-based DDoS attacks

DDoS attacks are among the most prevalent attack types that can be initiated across all layers of IoT architecture. These attacks target various layers, including the widely used RPL network layer protocol, transport layer protocols like TCP, and protocols of the application layer such as AMQP, CoAP, SNMP, and MQTT [30]. Recently, DDoS attacks have been increasing in both size and complexity, primarily due to the vulnerabilities present in resource-constrained IoT devices lacking adequate protection [31]. Additionally, DDoS attacks often serve as a precursor to other unprecedented and intricate attacks on IoT systems. These attacks originated from the botnet of vulnerable IoT devices in recent years as mentioned in the trends and growth of DDoS attacks due to various IoT Botnet families like Mirai, Meris, Fodcha, etc. [7,32]. An IoT botnet is an accumulation of compromised IoT computing devices, which have been taken over by malicious software, enabling remote control by a “botmaster” or “botherder”. The botmaster can then issue instructions to this network of compromised bots, directing them to carry out various tasks, which may include launching attacks like distributed denial of service (DDoS), distributing spam, or simply propagating and infecting additional devices [33]. In the diverse IoT landscape, malicious actors have identified a vulnerability resulting from numerous manufacturers prioritizing price and time-to-market over security when developing their devices. Firmware updates might not be accessible or managed effectively, and there is often inadequate communication about them. Additionally, default credentials are frequently employed, rendering IoT devices a tempting target for malicious actors. The main targets of IoT botnets are DDoS attacks, spam emails, Malware spreading, etc. However, the maximum utilization of IoT botnets is for launching DDoS attacks which is the most stealthiest threat in the IoT market [34].

6.1. Life cycle of botnet in IoT networks

An IoT botnet typically undergoes a series of stages to compromise IoT devices and manipulate them to serve the botmaster's purposes. The specific number and names of these stages may vary. However, the wide-ranging description of the botnet life cycle is shown in the following Fig. 6.

6.1.1. Scanning IoT devices stage 1

Scanning serves as the initial phase in the lifecycle of a botnet. During this stage, the attacker or botmaster conducts a thorough scan of the target network or devices connected to gather initial data regarding protocols, services, operating systems, and other relevant information. Attackers employ various techniques for scanning target devices and IoT networks. Popular scanning tools commonly utilized for this purpose encompass NMap, Shodan, ZMap, Masscan, OpenVAS, IoTSeeker, Synk, etc. [35].

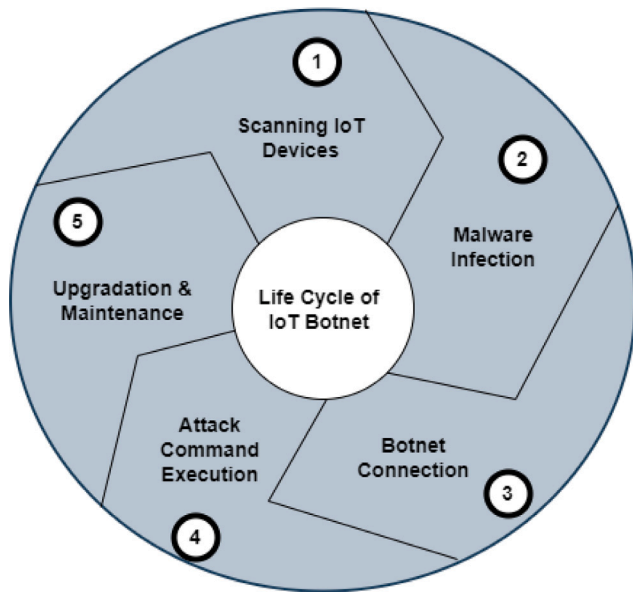


Fig. 6. Life cycle of a typical botnet in an IoT network.

6.1.2. Malware infection stage 2

Following the reconnaissance phase in which the attacker or bot master collects information about devices and the target network, stage 2 includes different exploitation techniques to uncover vulnerabilities. Successful exploitation permits the injection of malware into the target, often accomplished through phishing or email attachments [35]. Once initiated, a shell code is installed on the victim's device, gathering additional information and transmitting it to a command-and-control server. Subsequently, a bot program (malware) is downloaded and configured based on the target device's characteristics, marking the secondary injection phase. The infected device is then referred to as a 'zombie'.

6.1.3. Botnet connection stage 3

Upon activation, the bot program initiates communication with the Command and Control (C&C) server or the botmaster to confirm its status as a legitimate bot. The initial connection efforts by the bot to the C&C server for receiving directives from the botmaster are referred to as a rally. Subsequently, the injected malware executes according to the attacker's design, potentially functioning as a Trojan that responds to specific triggers. This running malware establishes a link between the target machine and the botmaster, facilitating the reception of further instructions.

6.1.4. Command execution stage 4

Upon establishing a connection with the botmaster or Command and Control server, the infected device becomes a part of the botnet. It enters a waiting phase, awaiting commands from the Command and Control server to carry out malicious tasks directed by the botmaster. These tasks encompass scanning for new bots, spam distribution, and executing DoS attacks. When numerous infected devices are connected to the botmaster, commands are dispatched to orchestrate flooding or DDoS attacks against a target server [35].

6.1.5. Upgradation and maintenance stage 5

To ensure continued stealth within the compromised system, a bot program residing in a compromised IoT device must undergo periodic updates and maintenance. This upkeep is crucial for an attacker to sustain the malware infection, enabling the utilization of infected devices in future malicious attacks.

6.2. IoT botnet families/variants

Initially, we provide detailed information on the prominent IoT malware capable of DDoS attacks over the past few years, categorizing them into families sharing common features and characteristics. Our focus is exclusively directed towards IoT malware that possesses the capability to launch DDoS attacks stemming from IoT devices, deliberately excluding IoT malware with distinct objectives.

6.2.1. Linux.Hydra

Linux.Hydra is an open-source tool that was introduced in 2008 specifically targeting routing devices employing MIPS architecture. The exploitation phase depends on a dictionary attack or, in the case of D-Link routers, exploits a well-known vulnerability of authentication. Once a device is compromised, it integrates as a part of an IRC-based network and can execute a SYN Flood attack. The malware's documentation mentions Linux.Hydra's ability to launch a UDP Flood attack, but this capability is not evident in online sources. Although relatively straightforward, this malware laid the foundation for subsequent MIPS-targeting malware [36].

6.2.2. Psyb0t

This malware appeared in 2009. In contrast to its predecessor, Psyb0t possesses the capability to execute ICMP and UDP Flood attacks [37]. It targets the MIPS architecture most of the time, which encompasses network devices. Although Psyb0t's source code has not been revealed, the two malware share numerous similarities, leading to the reasonable assumption that Psyb0t is a descendant of Linux.Hydra [38].

6.2.3. BASHLITE/Lizkebab/Torlus/Gafgyt

In 2014, the prevalent BASHLITE malware displayed resemblances to the Spike malware family. It utilized a streamlined IRC communication protocol, extensively modified to create a botnet architecture independent of IRC servers, categorizing it as Agent-Handler-based, not IRC-based. It exhibited an impressive range of vulnerable architectures, including SPARC devices. The DDoS attacks it executed were rudimentary, comprising traditional ACK, UDP, and SYN Flood attacks [39].

6.2.4. Mirai

Emerging in 2016, this malware stands as one of the most threatening IoT-based DDoS attacks in recent years, significantly altering perceptions of IoT security. It orchestrated the largest DDoS attack in history, creating a massive Agent-Handler botnet by compromising vulnerable IoT devices through a basic dictionary attack. This malware can execute a diverse range of DDoS attacks using different protocols (e.g., UDP, TCP, and HTTP) and exploit devices across various architectures. It is still among the most dangerous and dreadful DDoS-capable malware of IoT [39].

6.2.5. 3ve

Comprising three interlinked sub-operations, each specializing in ad fraud and evading detection, 3ve orchestrated extensive fraudulent ad bid requests and generated millions of deceptive domains. In 2018, a joint investigation by Google, White Ops, and law enforcement agencies led to an unprecedented dismantling of the botnet. Over two years, the botnet network grew from modest origins to include numerous IP addresses in residential and business domains [18]. The capacity of 3ve to independently establish a botnet by producing fake copies, hiding IP addresses behind proxies, and tampering with the Border Gateway Protocol (BGP) to trick advertisers and optimize earnings was what distinguished it from earlier botnets [17].

6.2.6. Wirex

In August 2017, widespread distributed denial of service attacks targeted against hospitals, casinos, and domain registrars revealed the existence of the WireX botnet. The command-and-control protocol used a particular delimiter string. Along with many Android smartphones, it had malware that was disguising itself as legitimate apps. Once these applications have gained access to a device, they wait for attack orders from a command-and-control server. These apps were primarily multimedia players, memory managers, or widgets. These apps used Android devices while operating in the background and launched attacks [18,40].

6.2.7. Reaper

While Mirai primarily exploited default or weak passwords to compromise network routers and IP cameras, a new botnet threat, also referred to as Reaper has emerged. Reaper employs actual application hacking techniques to gain control over computers [18]. This malware has amalgamated various hacking techniques of IoT, executing nine attacks on routers from D-Link, Netgear, and also digital surveillance systems produced by companies like AVTech, Vacion, and GoAhead [41].

6.2.8. Torii

Dr. Bontchev detected Torii using 'Tor' exit nodes in his honeypot, hence the name 'Torii'. This malware could target a wide range of devices, including current systems, smartphones, and tablets using various architectural models like MIPS, ARM, x86(64-bit), and more. Torii seeks Telnet ports to exploit weak device authentication in the same way as done by Mirai. However, it distinguishes itself as a more sophisticated IoT malware as it can transfer the appropriate payload for infecting similar models [17].

6.2.9. Meris botnet

Meris represents a novel DDoS botnet variant, encompassing approximately 30,000 compromised systems and devices. It is also known as "Plague" in Latvian. This botnet is assembled from sophisticated devices that necessitate Ethernet connections for operation. Notable features of Meris in orchestrating attacks include its utilization of HTTP Pipelining and a proxy. The botnet floods the servers as part of its strategy to execute large-scale DDoS attacks [42].

7. Launching methodology of an IoT-based DDoS attack

DDoS attackers intrude on IoT networks by exploiting the limitations of network resources, including restricted network links, battery power, storage capacity, and network bandwidth, and generate an army of compromised IoT devices like cameras, sensors, meters, thermostats, RFID tags, etc which are soft targets by the aggressor or cyber criminals and added to IoT botnets. After that, the cyber criminals overwhelm the target server or the network resources by flooding it from different locations using a botnet of IoT devices. Then, the targeted device or server refuses to fulfill client requests because of an overwhelming influx of data needlessly consuming all available bandwidth. Consequently, the target servers become inaccessible for additional requests. This data overflow is deliberately directed at the servers by the attacker in the guise of fraudulent requests. As a consequence, even legitimate requests face disruptions and are prematurely discarded. The outcome of such an infringement in a distributed setting, wherein data transfer occurs among various network devices dispersed across the network without central control, is commonly referred to as a Distributed Denial of Service Attack [19] as illustrated in Fig. 7.

8. Taxonomy of various IoT-based DDoS attacks

IoT attackers aim to exploit the security vulnerabilities and infiltrate systems. Their goal is multi-dimensional and hence it is not limited to a single layer or security principles. DDoS attacks exhibit a broader range of diversity and intricate due to heterogeneity of IoT devices. All such attacks can be further classified in terms of different IoT Layers. An essential point to note regarding DDoS attack classification is its dependency on the impact within an IoT network's server site. DDoS attack categories resemble traditional ones but adapt to the network architecture as a reference model. Another DDoS class emerges, centered on generating numerous pseudo requests from IoT sources to target servers. Currently, the sole known method to generate this volume of fraudulent requests via the IoT network involves the use of bots and malware [4]. Here we are classifying the DDoS attacks at various layers of the IoT network as shown in Fig. 8. Every layer can be specifically targeted to compromise the security of an IoT network.

8.1. Application layer attacks

The attackers aim is to breach the application layer of the IoT network infrastructure by overwhelming the application or web server with an excessive rate of requests per second. Detecting and mitigating these attacks presents a challenge because they typically produce traffic at a slower rate, and the generated requests seem authentic [43]. However, these legitimate requests initiate backend processes that render resources inaccessible. Various DDoS attacks on the application layer are as follows:

8.1.1. HTTP/HTTPS floods

Attackers flood the IoT application server with an overwhelming number of HTTP or HTTPS requests. This can exhaust server resources, making the application unresponsive.

8.1.2. DNS reflection attacks

Attackers exploit insecure DNS servers to amplify their attacks on IoT applications. They send DNS queries with the IP of the victim as the source, causing a flood of responses to overwhelm the server.

8.1.3. Slowloris attacks

These attacks involve opening numerous connections to the IoT application server and sending partial HTTP requests. By keeping connections open, attackers consume server resources, eventually causing it to become unresponsive.

8.1.4. Application layer protocol exploits

Attackers target vulnerabilities within specific application layer protocols used in IoT devices, such as MQTT or CoAP, to disrupt their functionality.

8.1.5. API abuse

Attackers misuse APIs (Application Programming Interfaces) exposed by IoT applications, sending excessive or malicious requests to destabilize the application's performance.

8.2. Transport and network layer attacks

The attacks at the transport layer and Network layer are also known as infrastructure layer attacks. the infrastructure layer of IoT encompasses the underlying technologies and components required to establish reliable and efficient connectivity, data transfer, and communication within IoT networks. Attacks at this layer are further classified into two categories, volume-based attacks and protocol-based [44]. Typically, these attackers utilize reflection or amplification methods to launch their attacks. In reflection, the attacker utilizes IP address spoofing to bounce the sent request back as an unintended response to the victim, causing congestion in the network of the victim [43]. Amplification generates larger responses for smaller requests, needlessly consuming bandwidth [4].

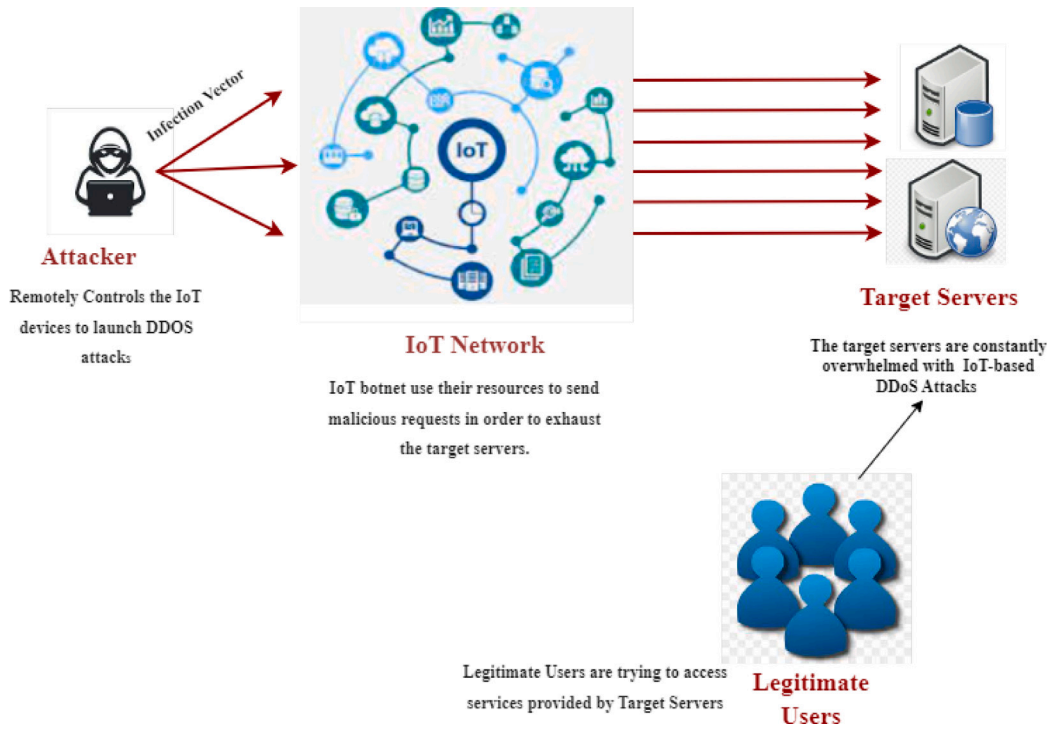


Fig. 7. Launching mechanism of an IoT-based DDoS attack.

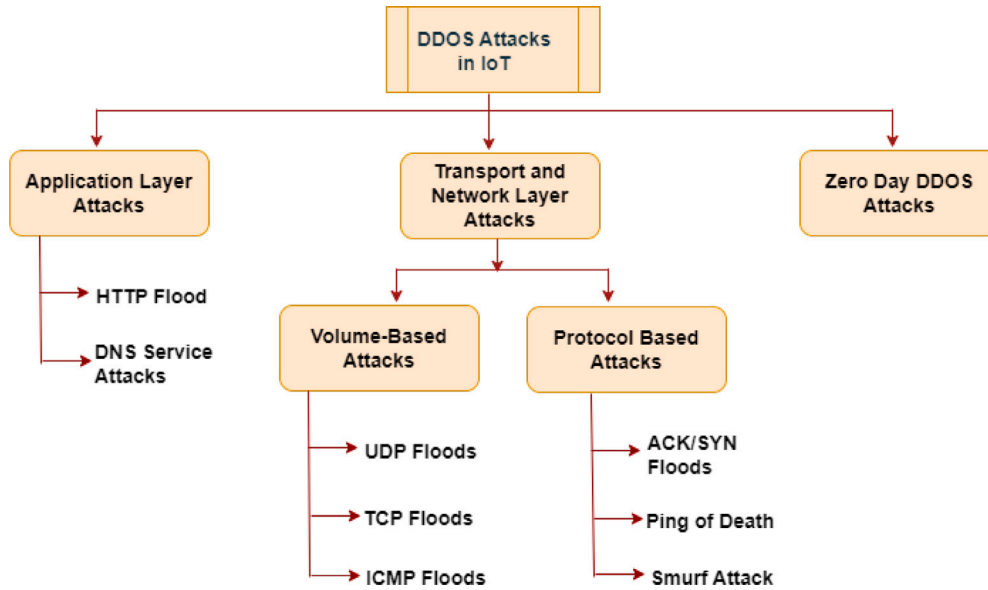


Fig. 8. DDoS attacks taxonomy in IoT layered architecture.

8.2.1. Protocol-based attacks

Protocol-based attacks are also referred as Resource Depletion attacks and are accountable for depleting not only server resources but also intermediary communication devices like load balancers and firewalls. These attacks are quantified in terms of packets per second. Illustrations include SYN floods, Smurf DDoS, fragmented packet attacks, and more [43].

8.2.2. Volumetric attacks/bandwidth depletion attacks

Volumetric attacker or Bandwidth Depletion attacks, overload the target system's bandwidth by generating an enormous traffic load measured in bits per second. They are straightforward to execute, utilizing amplification and reflection techniques for the assault. Research

has shown that as much as 65% of attacks fall into the category of volume-based attacks. Its examples are ICMP floods, UDP floods, TCP floods [17].

The following DDoS attacks have become increasingly prevalent recently:

8.2.3. ACK and SYN flood attack

A TCP three-way handshake process is used for establishing connections in an ACK Flood attack. It involves sending ACK-enabled packets with forged source addresses to the target device, which drops these packets as they lack a genuine connection. However, processing each incoming fraudulent packet depletes the target's resources, potentially causing unresponsiveness [4]. A SYN Flood attack occurs when a hacker

bombards a target server with a large number of TCP SYN packets, attempting to establish a connection but failing to complete the three-way handshake. This overwhelms the server, exhausting network resources and causing service disruption [17].

8.2.4. Domain name server amplification attack

A hacker or cyber-criminal exploits open DNS resolvers by transmitting DNS lookup requests with falsified source addresses to these resolvers in a DNS amplification attack [43]. In response, the DNS resolver sends a DNS record response, but it mistakenly directs this larger response to the target device in place of the attacker's device. This response typically contains extensive information records for a domain, making it significantly larger than the initial query, in which only the domain name is included [45].

8.2.5. NTP amplification attack

The NTP (Network Time Protocol) is employed to align local and global time over the internet [43]. Nevertheless, it can be exploited for DDoS attacks through query requests with falsified source addresses. These queries result in significant replies, such as lists of the servers 600 most recent connected hosts, which are needed to calculate UTC. Frequent generation of these larger responses leads to flooding, causing the server to reject subsequent requests.

8.2.6. UDP fragment attack

The intruder sends oversized packets meant to fragment and reassemble at the destination in UDP fragment attack. The repeated failures in reassembly and fragmentation overload the device, causing the server to reject other packets. Termed a "Ping of Death" attack, it involves sending multiple malformed pings that, when reassembled after fragmentation, exceed the maximum IP packet length at the data link layer, leading to memory buffer overflow and a DOS attack [43].

8.2.7. UDP flood attack

A large volume of UDP packets is maliciously sent to a target server in UDP (User Datagram Protocol) flood attack. Unlike TCP, UDP lacks the handshake mechanism, making it susceptible to exploitation. The attacker overwhelms the target's resources, causing service disruption or unavailability.

8.2.8. HTTP/HTTPS flood attacks

An HTTP/HTTPS flood attack is one of the categories of Distributed Denial of Service attacks that overwhelm web servers with a high volume of HTTP or HTTPS requests. This influx of traffic exhausts server resources, rendering the web service completely unavailable or slow to legitimate users.

8.2.9. ICMP flood attacks

ICMP (Internet Control Message Protocol) typically handles the generation of error messages to report network or destination failures, such as data buffering issues at gateways or packet unreachability. ICMP includes the "Ping" function, which triggers an echo request and expects an echo reply. Failure to receive this reply signifies an inactive or non-ping-capable host. In such cases, echo requests are sent incessantly without awaiting replies, needlessly inundating the network with traffic, a phenomenon referred to as a "smurf attack".

8.2.10. Zero-day DDoS attacks

DDoS attacks that exploit previously unknown vulnerabilities or flaws in IoT devices, networks, or protocols are known as Zero-day DDoS attacks in the context of IoT networks [46]. These attacks target IoT systems before developers or security experts have had the opportunity to discover and address these vulnerabilities. Zero-day IoT-based DDoS attacks are particularly concerning because IoT devices frequently have insufficient security policies in place, making them susceptible to exploitation. These attacks can result in the disruption of IoT services and potentially compromise the privacy and security of IoT devices and data.

Table 3

Prominent tools used to launch DDoS attacks.

Sr. No.	Tool	Type of DDoS attacks
1.	LOIC	TCP flood, HTTP flood, UDP flood
2.	HOIC	HTTP flood
3.	Hping	TCP flood, ICMP flood
4.	#RefRef	On database
5.	Slowloris	Exhaust Server
6.	RUDY	HTTP Flood
7.	Golden Eye	HTTP GET/POST
8.	XOIC	TCP, UDP, ICMP, and HTTP
9.	Tor's Hammer	HTTP POST
10.	DAVOSET	HTTP Flood

9. DDoS attack launching software tools

Various techniques exist for executing DDoS attacks, encompassing both automated and manual approaches. Manual attacks rely on the attacker's motivation and skills, with IoT environments commonly exploited to form botnets. IoT devices themselves can serve as potent attack tools. Nevertheless, certain scanning and DDoS generation tools are employed, although necessitate a basic understanding for effective utilization. These tools continually evolve to meet shifting attacker demands, evolving to possess enhanced sophistication, stealthiness, and greater attack potency. Here are some tools employed for DDoS attacks [37,47–49] as mentioned in Table 3.

- **LOIC (Low Orbit ion Cannon)**- LOIC is an open-source tool, crafted by Praetox Technology [16], that aids developers in generating significant network traffic for high-load server testing. Regrettably, Anonymous repurposed it for various disruptive flooding attacks, utilizing features like Hive-mind for more potent DDoS assaults. It sends TCP, HTTP, and UDP requests to the server.
- **HOIC (High Orbit ion cannon)**- The HOIC tool gained popularity among attackers who sought anonymity, as it improved upon LOIC's limitations. With a cross-platform script and intuitive GUI, HOIC makes HTTP POST and GET requests simpler. Booster scripts, and basic text files interpreted by key applications, enhance attack efficiency [47].
- **Hping**- This fundamental command tool is employed for generating substantial volumes of TCP traffic. It closely resembles the function of a ping but boasts additional capabilities. With this tool, it becomes possible to spoof source IP addresses and transmit ICMP and TCP request packets.
- **#RefRef**— This tool specializes in pinpointing vulnerabilities within a server's SQL backend. It streamlines attacks, requiring only a single machine when targeting a susceptible SQL backend, executing in 17 to 20 s with an impact lasting roughly 40 min [50].
- **Slowloris**- Slow and low DDoS attacks, challenging to detect, fall into RSnake's purview and are orchestrated by grey hat hackers. These attacks fragment requests to deliberately prolong server response times, depleting server resources [51].
- **RUDY**— Slowloris employs prolonged HTTP headers for slow-rate DDoS attacks, whereas RUDY utilizes lengthy HTTP POST submissions with a single byte injection into the application's POST field, enabling multiple simultaneous connections to the target server [16].
- **Golden Eye**- A multi-threaded attack tool designed that uses HTTP GET and POST requests to initiate DDoS attacks. While lacking IP spoofing capabilities, it functions on widely used operating systems like Windows, Linux, and MAC [50].
- **XOIC**- A graphical interface tool employed for initiating UDP, TCP, and ICMP packets, allowing users to specify IP addresses, ports, and protocols manually. Developed in C#, it is regarded as more potent than the LOIC tool [16].

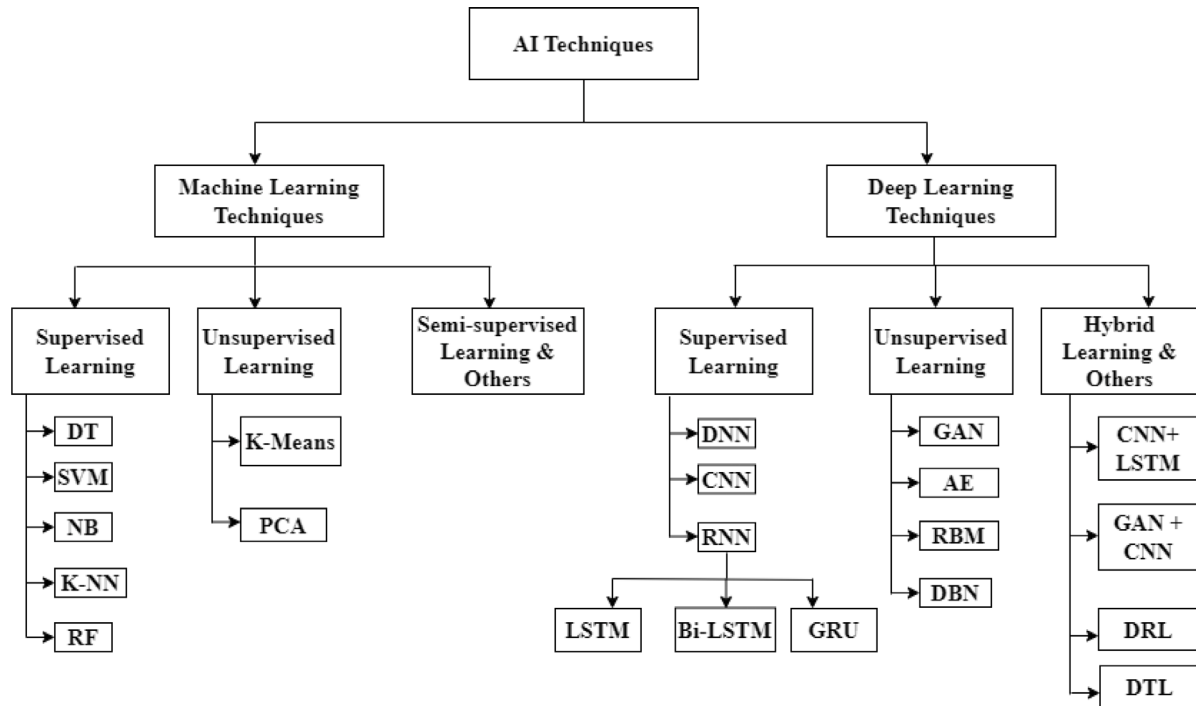


Fig. 9. Taxonomy of AI techniques.

- **Tor's Hammer**- Tor's Hammer is a Python-based attack tool designed for launching HTTP attacks. This tool operates via TOR networks and depletes the victim's server resources [43].
- **DAVOSET** – DAVOSET is a software which is employed to launch a DDoS attack by exploiting the functionality of any website. It provides a command line interface to perform DDoS attacks which offers support for cookies and can also be employed to execute attacks utilizing XML external entities [52].

10. Taxonomy of AI techniques for IoT security

Artificial intelligence is capable of simulating the human mind and thought processes. AI is not the same as human intelligence, yet AI can think more intelligently than humans [53]. Various cybersecurity professionals are increasingly moving towards Artificial Intelligence (AI) to safeguard the systems against cyber threats. Artificial Intelligence is employed for IoT security, where it analyzes traffic patterns to identify activities characteristic of a potential attack [54]. The development of IoT technologies has brought a great attention to AI. These are helpful for securing the IoT networks from harmful attackers in terms of finding out the anomalous behavior of traffic patterns. Machine Learning (ML) and Deep Learning (DL) are the two main branches of AI as shown in Fig. 9 are described below:

10.1. Machine learning techniques

Machine learning is a growing field in the domain of data science. It is used to train the algorithms for predictions and classification. The taxonomy of ML techniques for IoT-based DDoS attack detection contains three primary categories that are supervised learning, unsupervised learning, and semi-supervised learning [55].

10.1.1. Supervised learning

Supervised learning uses the labeled features for the purpose of training the model. Various techniques used in supervised learning include Decision Tree (DT), Naive Bayes (NB), Support Vector Machine (SVM), Random Forest (RF), K-nearest neighbour (kNN) which are illustrated as below:

- **Decision Tree**: The Decision Tree based technique is used to design a prediction model that learn from the labeled dataset. The labeled data is represented in the form of branches and leaves of the decision tree. It is a transparent and easy-to-use method. It needs large storage if a number of decision tree grows [55]. Samples are classified with the help of conditions in the decision trees and provide high accuracy of attack detection in case of known traffic patterns, but it is not suitable for anomalous traffic patterns.
- **Support Vector Machine**: Support Vector Machine is a robust supervised machine learning technique employed for tasks involving both classification and regression. Support vector machine demonstrates notable effectiveness, especially when dealing with data that is not linearly separable, requiring transformation into a higher-dimensional space to identify a separating hyperplane. In the case of classification, the objective of the SVM technique is to identify the optimal hyperplane that best divides the data points of different classes in a high-dimensional feature space [56]. "The hyperplane is chosen in such a way that it maximizes the margin between the two classes. The margin is defined as the distance between the hyperplane and the closest data points of each class. These closest data points are called support vectors" [57]. The main strength of the SVM lies in its ability to handle complex, nonlinear data by utilizing the trick of kernel that implicitly maps the original data points into a higher-dimensional feature space, where they become more likely to be linearly separable. Common kernel functions are the Polynomial Kernel, Linear Kernel, and the Radial Basis Function (RBF) Kernel [58].
- **K-Nearest Neighbor**: The K-Nearest Neighbors (kNN) algorithm is a simple and popular machine learning technique used for both classification and regression tasks [59]. It is a type of instance-based or lazy learning algorithm, meaning that it does not explicitly learn a model during the training phase. Instead, it stores the training data and makes predictions based on the proximity of new data points to the existing ones [60]. The KNN algorithm enables WSN nodes to make decisions based on the consensus of their nearest neighbors, thereby providing a distributed and collaborative blackhole detection mechanism. The collective intelligence of the network helps in effectively identifying

and mitigating potential blackhole attacks, increasing the overall reliability and security of the wireless sensor network. kNN aggregates the votes received from its K nearest neighbor's attack vectors. If the majority of the neighbors indicate a direct path to the base attack, the node assumes it is safe to forward the data through this path. However, if there is no majority or the majority indicates no direct path, the node suspects the presence of a blackhole in the vicinity [61].

- **Naive Bayes:** Naive Bayes (NB) is a widely used supervised classifier of machine learning that relies on Bayes's Theorem. The naive Bayes classifier operates under the assumption that every characteristic in the input data is unrelated to every other feature, which is frequently untrue in practical applications. Nonetheless, the naive Bayes classifier is popular because to its effectiveness and strong performance in several real-world applications, even with this oversimplifying premise. It can be used to detect attack for both binary class and multi-class attack classification in cyber security [62].
- **Random Forest:** The Random Forest Classifier is very flexible and easy to use, making it suitable for solving issues involving both regression and classification. The method is a useful tool for a variety of machine learning prediction tasks because of its strength in handling complicated datasets and mitigating overfitting issues. The Random Forest classifier can handle data sets with both categorical variables and continuous variables. RF gives better results in the case of regression and classification. Diversity, dimensionality reduction, and stability are its main features [63].

10.1.2. Unsupervised learning

Unsupervised learning analyzes and groups unlabeled datasets using machine learning techniques. These algorithms find hidden patterns in the data without requiring human assistance [64]. The most commonly used techniques under unsupervised learning are described below:

- **K-Means Clustering:** K-Means clustering is a technique of unsupervised ML approach. The number k denotes the number of clusters that the algorithm is expected to produce [65]. The aim of this approach is to identify clusters in the data. The procedure is iteratively allocating every data point to one of the clusters according to the given attributes. There will be samples with comparable characteristics in each cluster. To arrive at a final conclusion, the k-means method uses repeated refining. "The dataset, which includes a set of characteristics for each sample in the dataset, and the number of clusters are the algorithm's inputs" [66]. Each sample is matched to the closest cluster centroid using the squared Euclidean distance once the k centroids have been initially computed. The centroids of a cluster are once more determined when all the data have been allocated to a certain cluster. This method repeats these steps until no such data sample that can change the clusters exists [67,68]. This approach is used to detect the anomalous behavior of traffic patterns.
- **Principal Component Analysis (PCA):** PCA is a dimensionality reduction technique while retaining the majority of the information contained in the original large dataset [66]. This approach converts the correlated features with less number of uncorrelated features. This is the primary concept of the principle component. This principle is used for securing IoT systems from real-time attack detection [69].

10.1.3. Semi-supervised learning

Semi-supervised learning is a hybrid technique of both supervised learning and unsupervised learning [70]. Therefore, It trains a machine learning classifier using both labeled and unlabeled data [66]. This approach is not successful in achieving attack detection in comparison to supervised learning. Thus, semi-supervised learning technique is utilized by very few studies for the security of IoT systems.

10.2. Deep learning techniques

Deep learning is a machine learning sub-field. The design and development of a neural network for analytical reasoning that mimics the functioning of the human brain. It simulates how the human brain processes information such as sounds, words, and visuals [53]. DL techniques are further categorized into supervised learning and unsupervised learning. In deep learning techniques, features are automatically generated whereas features are extracted manually in machine learning techniques [53]. The various categories of DL techniques are described below:

10.2.1. Supervised learning

Supervised learning uses the labeled dataset for training purposes. Various methods that are included in this category are as follows:

- **Deep Neural Network(DNN):** A deep neural network is also referred to as an artificial neural network (ANN) which contains input, output, and more than two hidden layers [71]. DNN is the fundamental building block of deep learning. It acts similarly to the human brain which contains billions of neurons that are used for computation. In the initial stage, DNN is used to find the patterns based on the given input, and its output is compared with the actual output. The difference between the calculated output and actual output is known as an error [43]. In the next step, the value of weight and bias is adjusted to minimize the error through backpropagation. During the process of backpropagation, the difference between the calculated output and the actual output gives the least error. In that case, the designed model gives the accurate performance [72].
- **Convolutional Neural Network(CNN):** The CNN comprises four layers, namely the convolutional layer, pooling layer, flattening layer, and fully connected layer. The major computation occurs at convolutional layer in which filters are applied to get the feature map [73]. Next, conditionality reduction of the feature map is done in the pooling layer with the help of max pooling or average pooling [74]. It is used for down sampling. The output of the pooling layer received in multi-dimensional data which is converted to 1-D array that becomes the input for the fully connected layer. Then, 1-D array data is used for classification task on the basis of extracted features in different layers [73,75].
- **Recurrent Neural Network(RNN):** The RNN approach is utilized to process the sequence data which is not handled by the traditional layers of DNN and CNN [53]. By maintaining the prior data, RNN can predict the next word in a phrase by providing the current step with both the input and the output from the previous step. However, RNN face the problem of gradient vanishing [76]. The capability of efficiently handling sequential data by RNN approach in threat detection. It is a useful approach for IoT security to handle bid data with sequential features. It can categorize IoT network traffic into either malicious or benign traffic [53].
 - (a) **Long short-term memory (LSTM):** The limitation of RNN as a vanishing gradient is solved by Long short-term memory. It consists of various memory blocks or gates. Memory blocks utilize three processes known as gates—forget, input, and output gates—to choose which information to save or discard that allows it to effectively model and learn complex temporal patterns [75]. LSTMs have proven to be particularly effective in tasks such as time series prediction, speech recognition, and language translation [77]. LSTM networks are also used to analyze sequences of network packets and detect patterns of IoT-based DDoS attacks.
 - (b) **Bi-Long short-term memory (Bi-LSTM):** Bidirectional-LSTM is an extension of the traditional Long Short-Term Memory (LSTM) approach, designed to enhance the modeling of sequential data by considering information from both past and future time steps. While standard LSTMs only capture dependencies from the past,

Bi-LSTMs process input sequences in both forward and backward directions [78]. Bi-LSTMs can capture complex patterns and dependencies in sequential data due to this bidirectional processing. This technique is beneficial for those activities that are crucial from both past and future contexts for accurate predictions or classifications. However, the model designed through Bi-LSTM requires more time for training in comparison to tradition LSTM [79].

(c) *Gated recurrent unit (GRU)*: Gated Recurrent Unit is a simplified version of LSTM. It has a few number of parameters in comparison to LSTM [80]. GRU integrates the forget and input gates into an “update gate” and also merges the hidden and cell states with a few other modifications [81].

10.2.2. Unsupervised learning

Unsupervised deep learning is also known as generative learning. Various commonly used unsupervised DL approaches are Generative Adversarial Networks (GANs), Deep autoencoders (AEs), restricted Boltzmann machines (RBMs), deep belief networks (DBN) etc.

10.2.2.1. Deep autoencoders (AEs). Deep autoencoders are used for the process of extracting features. When used for representation learning, Autoencoders (AEs) can decrease dimensionality without the requirement of any prior knowledge of the data by learning features instead of manually constructed features as in machine learning. AEs need a significant amount of processing time. The learning model using AEs becomes complicated if the training dataset is not representative of the testing dataset [66].

10.2.2.2. Restricted Boltzmann Machines (RBMs). Restricted Boltzmann Machines (RBMs) are a type of generative stochastic artificial neural network [82] that plays a pivotal role in the domain of deep learning. It comprises two layers: a visible layer and a hidden layer. Nodes within a same layer are fully connected, but there are no connections between nodes within the same layer. RBMs are used for extracting various important features [66].

10.2.2.3. Deep Belief Networks (DBN). A Deep Belief Network (DBN) is a type of generative model that employs deep neural networks. DBNs include stacked RBMs that run greedy layer-wise training to achieve strong performance in an unsupervised environment. These networks incur more computation cost due to lengthy initialization procedures [66]. These are also used for malicious attack detection [83].

10.2.3. Hybrid learning

Hybrid learning utilizes the learning models by combining any two techniques i.e. supervised learning or unsupervised deep learning. It covers the strengths of both learning approaches to improve the performance parameters of the model. Hybrid learning is also referred to as deep ensemble learning or deep transfer learning. Various examples of hybrid learning are CNN+LSTM, GAN+CNN, etc. The main goal of hybrid learning is to create more robust, adaptable, and efficient models.

11. State-of-the-art IoT-based DDoS attack detection AI techniques

An extensive literature has been examined related to the detection of IoT-based DDoS attacks which use the machine learning (ML) and deep learning (DL) techniques. Furthermore, we have illustrated and comparatively analyzed the performances of various machine learning and deep learning classifiers being used by the reviewed studies. Some of the studies have also utilized hybrid techniques that are capable of detecting DDoS attacks in the IoT ecosystem. The comparative analysis of the studies is shown in Table 4

Soe et al. [84] developed a deep learning model for identifying DDoS attacks in the IoT ecosystem by using an ANN (Artificial neural networks) classifier on the publicly available dataset BoT-IoT. Authors

have used SMOTE as a re-sampling technique for balancing the various instances of the dataset. The model accuracy has been improved by using the re-sampling approach.

The study [85] has provided an intrusion detection system (IDS) to detect DDoS attacks by using the integration of deep learning technique CNN and LSTM with multi-objective optimization technique NSGA-II for feature selection. The designed model is trained and tested on the CISIDS2017 dataset with an accuracy of 99.03% and compared with existing state-of-the-art machine learning techniques that show its high performance.

The research [86] suggests a time-stamped bi-directional model that utilizes the deep learning technique gated recurrent unit for detection and classification of distributed denial of service attacks in IoT networks with favorable detection performance. It is experimented with using two datasets, one is the publicly available dataset BoT-IoT dataset with traffic of IoT devices and the other one is the private dataset NTHU which is street lamped dataset collected from real IoT networks.

Mubarakali et al. [87] proposed an SDN model that employs the SVM algorithm to detect DDoS assaults. In this framework, the SDN controller gathers the date of flow table status and the SVM algorithm accesses the traffic and detects the IoT-based DDoS attacks in multi-class attack classification. Through the deployment of the SDN environment, the authors examine the variations in feature values of the traffic data. The average accuracy attained by the proposed model is 96.23% with a high detection rate (DR) and low false alarm rate (FAR).

The research [88] focuses on defensive techniques against IoT DDoS assaults and suggests FlowGuard, an IoT edge-centric defense system, for the detection, identification, and mitigation of IoT-based DDoS attacks using deep learning techniques Convolutional Neural Networks and Long short-term memory on the synthetic datasets using DDoS simulators BoNeSi and SlowHTTPTest, and integrate it with the CICCDDoS2019 dataset. The model is deployed in edge servers that meet the delay requirement of IoT devices and outperformed when compared with the existing studies. However, the model needs high-speed processing units and memories.

Aysa et al. [89] illustrated the security flaws and vulnerabilities of IoT devices which are used as a weapon for the most dreadful DDoS attacks. The authors proposed four machine learning classifiers Linear support vector machine, J-48, Random Forest, and Neural Network to design a model on standard datasets. It was concluded in the study that the integration of the Decision Tree and Random Forest achieved more accuracy in detecting IoT-based DDoS attacks.

Chen et al. [90] introduced a multi-layer system to detect and mitigate DDoS attacks within IoT gateways by employing a Decision Tree as ML classifier. Experimental validation involves launching DDoS attacks in a real IoT scenario with eight smart poles and achieved F1-score exceeding 97%. The study also implemented the mitigation policy by getting the IP address and MAC address of malicious packets and forward it to SDN controller for blocking that malicious device. However, designed model is lacking the portability.

Vishwakarma and Jain [91] developed a honeypot-based real-time framework to detect Zero-Day distributed denial of service attacks (DDoS) in IoT networks by utilizing various ML classifiers decision tree, random forest, k-nearest neighbor, and support vector machine. However, the designed model has not been comparatively evaluated with the existing solutions to detect DDoS attacks in IoT networks.

Ravi and Shalinie [92] proposed a unique semi-supervised machine learning-based model to detect and mitigate the most harmful and stealthy DDoS attacks in IoT networks by utilizing two categories of datasets UNB-ISCX (Benchmark dataset) and Synthetic dataset by using their own testbed of IoT devices. The authors have achieved an accuracy of 96.28% and the proposed model also surpassed the performance of the existing state-of-the-art ML models.

Faruke et al. [93] have proposed an approach for DDoS attack detection in IoT networks which are becoming more prone to attacks

Table 4

Comparative analysis of DDoS attacks in IoT networks using ML and DL techniques between (2019–2023).

Study/ year of publish	Data set used	Attack classification	Technique	Proposed method	No. of features selected	Evaluation results	Limitations
[84] 2019	BoT-IoT	Binary	DL	ANN	41/43	Up to 100%	The dataset is highly imbalanced, Only Binary classification is considered.
[85] 2020	CISIDS2017	Binary	DL	CNN+LSTM	81/85	Accuracy = 99.03%	CICIDS2017 dataset is not balanced (No. of instances are reduced to make it balanced)
[86] 2020	1.BoT-IoT 2. Private Dataset	Binary	DL	Bi-GRU	21/43	Accuracy = 100%	No real-life implementation yet
[87] 2020	Synthetic	Multi-Class	ML	SVM		Accuracy = 96.23% High Detection Rate, Low False Alarm Rate	Normal Traffic is missing, Only Attack traffic has been introduced.
[88] 2020	Simulated + CICDDoS2019	Binary	DL	CNN, LSTM	10/83	Accuracy = 99.9% (CNN) 98.9% (LSTM)	Absence of multi-class classification of DDoS attacks
[89] 2020	Standard dataset	Binary	ML	LSVM, DT, RF, NN	40/115	Accuracy = 99.7%(DT)	The subcategories of attacks are missing. Not scalable to huge IoT Network.
[90] 2020	Synthetic	Binary	ML	Decision Tree		F1 score = 97% High Accuracy	Lack of model portability to new fields. Artificial marking of features.
[91] 2020	Synthetic dataset	Binary	ML	k-NN, RF, NN, DT, SVM	6	Accuracy = 99%	The Proposed model is not compared with any existing solutions.
[92] 2020	1. UNB-ISCX 2. Synthetic	Binary	ML	ELM	155	Accuracy = 96.28%	Scalability Issue
[93] 2021	CISIDS2017	Binary	DL	1D-CNN, MLP	38	Accuracy = 99.63% (CNN) 99.58%	Absence of testing the model in Real-Time IoT Network Traffic.
[94] 2021	Smart city simulated three datasets	Binary	DL	CNN, RBM	5	Accuracy = 99.5%	Absence of model application on Real Life smart city attack dataset.
[95] 2021	BoT - IoT UNSW-NB15	Binary /Multi Class	DL	LSTM	29/46	ACC = 99.529%	Need of high processing units for DL model
[96] 2022	CICDDoS2019	Binary	ML	k-NN, RF, DT, XGBoost	35 to 60	Accuracy = 98.34%	No Mitigation strategy
[97] 2022	BoT - IoT	Binary	ML/DL	DT, NB, KNN, RF, ANN	28/43	ACC = 96%	ML models have less accuracy than the DL model, The subcategory of attacks is not considered.
[98] 2022	BoT-IoT	Binary /Multi-Class	ML/DL	DT, RF, SVM RNN, LSTM, GRU, MLP	10/43	Accuracy more than 99%	Model performance needs to be checked on Real-Time traffic of the IoT Network.
[99] 2022	Banking Sector Dataset	Binary	ML	SVM, RF, k-NN	7	SVM is more robust	This model is limited to offline datasets, and may not work for real-time data sets.
[100] 2022	BoT-IoT	Multi-Class	ML/DL	DT, RF, KNN, MLP, RNN, LSTM	10/43	Accuracy = 99.81%	The model needs to be tested on multiple Datasets to check its performance.
[101] 2022	Simulated	Binary	ML	SVM, RF, LR, KNN, DT, NB	26	Accuracy = 99.2	Limited for binary class attack classification
[102] 2022	NSL-KDD	Binary	DL	RNN	10	Accuracy: 99.98%	The Dataset is obsolete.
[103] 2022	1. Synthetic 2. KDD-Cup	Binary /Multi Class	ML	k-NN, SVM LR, DT	2	Accuracy: 99.9% (DT)	ML classifiers may not give promising results on high quantity data generated by IoT Devices.
[104] 2023	1. BoT-IoT 2. TON-IoT	Binary	ML/DL	Linear SVM, NB, LR, ANN, CNN-LSTM	20/43	Accuracy = 99%	Passive IDS for SDN and intermediate Fog Layer of IoT networks.
[105] 2023	CICDDoS2019	Binary Class /Multi Class	ML/DL	ID3, NB, LR, RF, Ad Boost, CNN+LSTM	5	Precision 92%, Recall 89%, F1-score 90%	Hybrid DL classifiers are heavy and the model requires high computation power and processing time.

(continued on next page)

Table 4 (continued).

[106] 2023	BoT-IoT, CICIDS-2017, CICIDS-2018	Binary Class	DL	CNN	20	F1 Score = 98.94	High False alarm rate than supervised learning models and also, may have low performance in case of real-world environments.
[107] 2023	CICDDoS2019	Binary	DL	LSTM	71	Accuracy = 98%	The dataset is highly imbalanced.
[108] 2023	NSL-KDD	Binary	ML	LR, DT, k-NN, MLP	42/80	Accuracy = 99.9%	Performance may be different on the feature selection method. Also, a static threshold is used for Gini values.
[109] 2023	BoT-IoT	Binary	DL	LSTM, DBN, Bi-LSTM	26/43	Accuracy = 99.8%	Need of high processing Units
[110] 2023	Synthetic	Binary	ML	DT, SVM, NB	4	Accuracy = 97.4% 96.1%,98.1% Respectively	Multi-Class DDoS attack classification is missing, Only supervised ML classifiers are introduced.

as early as possible by utilizing two deep learning classifiers 1D Convolutional Neural Network and Multilayer Perception Method with the Random Forest method used as feature selector and implemented on publicly available dataset CICIDS2017 with accuracy 99.63% and 99.58% respectively. It is concluded that the proposed approach has better performance than the existing ML and DL models on a similar dataset.

Elsaedy et al. [94] have illustrated that DDoS attacks and replay attacks are harmful and destructive in smart cities due to the loss of human lives. The authors designed a hybrid DL model using the Convolutional Neural Network (CNN) and Restricted Boltzmann Machine (RBM) using three datasets smart river dataset, environmental dataset, and smart soil dataset. The accuracy achieved by using these datasets is 98.13%, 99.51%, and 98.37% respectively. The final evaluation results are also compared with the existing state-of-the-art ML and DL models.

Zeenshan et al. [95] have prepared a model to detect DoS and Distributed DoS attacks by merging the two publicly available datasets BoT-IoT and UNSWNB15 and also handled the issues of over-fitting and dataset imbalance. The authors have utilized the unsupervised deep learning technique Long short-term memory (LSTM) by selecting 26 features from both datasets and achieved an accuracy of 96.3%. In this model, both binary class and multi-class classification of attack categories are considered.

Gaur and Kumar [96] presented a hybrid methodology for the detection of IoT-based DDoS assaults using four machine learning classifiers Decision Tree, k-nearest neighbors, Random Forest, and XGBoost and three feature selection methods ANOVA, Extra Tree and chi-square have been employed for the development of machine learning model and achieved the average accuracy 98.34% for early detection Distributed Denial of Service attacks. First of all, the performance of the models has been evaluated using ML classifiers by selecting all feature sets. After that, various feature selection techniques are applied to reduce data dimensionality.

Kumar et al. [97] have used machine learning classifiers (Decision Tree, K-nearest Neighbor, Random Forest, Naive Bayes) and deep learning classifiers (Deep Neural Networks) to provide security solutions and analyze DoS and DDoS attacks by using BoT-IoT dataset as training dataset and also set up a testbed using 20 devices to generate a private dataset for testing purpose. They have attained an accuracy of 99.5% and 99.9% for DL and ML algorithms respectively. ML algorithms may not work with the same accuracy on huge data and resources.

Almaraz-Rivera et al. [98] have built an anomaly detection model by using machine learning and deep learning techniques for the detection of IoT-based DDoS attacks. The model has been implemented on the BoT-IoT dataset by selecting 10 features out of 46 and achieved an accuracy of more than 99%. The researchers have conducted a comprehensive experiment using three distinct feature sets for Binary and Multiclass classification of DDoS attacks (TCP, HTTP, and UDP). It

was concluded by the study that Multi-layer Perceptron and decision tree models are the best classifiers for identifying IoT-based DDoS attacks.

Islam et al. [99] have implemented an approach to find distributed denial-of-service (DDoS) attacks against financial institutions by utilizing the dataset of the banking sector. To boost the accuracy of generic models, additional complexity has been introduced to their architecture. They have used random forest algorithms (RF), K-Nearest Neighbors (KNN), and support vector machines (SVM) for the detection of denial-of-service (DDoS) assaults, the SVM demonstrates an accuracy of 99.5%, while the accuracy of KNN and RF are 97.5% and 98.74%, respectively. The comparison has shown that the SVM outperforms the RF, KNN, and state-of-the-art machine learning (ML) and deep learning (DL) techniques in terms of robustness.

Mihoub et al. [100] addressed an architecture having two components detection and mitigation against DDoS attack in IoT networks. The study explained the precise attack type and category of packet used in the attack. Several machine learning and deep learning classifiers DT, RF, MLP, KNN, RNN, and LSTM are used to classify the attack types as multi-class classifiers that follow the Looking-Back concept implemented using the BoT-IoT dataset in which 10 features out of 43 features are used. Random Forest classifier attains the maximum accuracy of 99.81% among all the ML/DL classifiers.

Gupta et al. [101] illustrated an approach to identify the DDoS attack traffic by employing various ML classifiers SVM, RF, LR, KNN, DT, and NB in IoT networks using the simulated IoT network to identify the benign and malicious traffic patterns and extracting the network flow dependent and flow independent 26 features. All the ML classifiers achieved an accuracy between 97% to 99% and concluded RF was the most resilient technique for detecting DDoS attacks. Furthermore, the study implemented a mitigation strategy and The assault traffic has been screened by the router at the local level.

Yousuf and Mir [102] proposed a novel algorithm for detecting Attack using a Live Capture Neural Network to identify IoT-based DDoS attacks using the deep learning technique RNN. The research has identified an EOpenDayLight as an SDN controller for attack detection. It was trained using the training dataset NSL-KDD. The proposed model DALCNN was also compared with other attack detection algorithms like MLP, SMO, j48, and IBK. The EOpenDayLight framework is also superior when these are compared with Ryu, Floodlight, ONOS, and OpenDayLight.

Ashafaq et al. [103] have classified IoT-based DDoS attacks by utilizing the four ML classifiers k-NN, LR, SVM, and DT. In this study, the authors have worked on two categories of models. In the first one, the dataset is synthetically generated using the XOIC tool and Wireshark by selecting only two features. In the second model, the KDD-Cup dataset is used by the author by selecting 10 features. After that, both the models were evaluated and it was concluded that the

Table 5

List of publicly available IoT datasets.

Dataset	Publication year	No. of instances	Malicious instances	Benign instances	No. of features	No. of IoT devices	Attack types
N-BaIoT	2018	70,62,606	65,06,674	5,55,932	115	9	BASHLITE (Scan, TCP, UDP, COMBO), Mirai (SCAN, SYN, UDP, UDPPPlain, ACK), Benign.
BoT-IoT	2019	7,33,60,900	7,33,51,357	9543	46	5 (Simulated)	Port Scanning, DoS/DDoS (TCP, UDP, HTTP), OS Fingerprinting, Data Theft, Keylogging, Benign
IoTID20	2020	6,25,783	5,85,710	40,073	86	Simulated	DOS, SCAN, ARP Spoofing, ACK Flooding, SYN Flooding, SCAN Port OS, HTTP Flooding, Scan Host Port, Benign
MedBioT	2020	3,74,33,560	54,77,468	3,19,56,092	100	83(Simulated + Real)	BASHLITE, MIRAI, Torii, Benign
IoT-23	2020	32,53,07,990	29,44,49,225	3,08,58,765			HTTP, DNS, DHCP, SSL, SSH, IRC
TON-IoT	2020	2,23,39,021	7,96,380	2,15,42,641	42	IoT/IIoT Devices	XSS, DoS, DDoS, Password Cracking Attacks, Reconnaissance, Ransomware, Backdoors, MITM, and Injection Attacks
Edge-IIoT	2022	2,09,52,648	97,28,708	1,12,23,940	61	10+ (IIoT and IoT Devices)	DDoS (TCP, UDP, HTTP, ICMP), Info Gathering (Port Scanning, Fingerprinting, Vulnerability Scanner), MITM, XSS, Uploading Backdoor Password, SQL Injection, Ransomware, Benign (14 attack types)

accuracy of decision tree classifiers is 99.9% which outperforms the other mentioned ML classifiers.

Khanday et al. [104] addressed a lightweight IDS for the protection of DDoS attacks in IoT networks using two datasets TON-IoT and BoT-IoT selecting the 20 best features from both datasets. SMOTE (Synthetic minority oversampling Technique) has been used to balance BoT-IoT datasets. After that model is trained by employing different machine learning and deep learning classifiers Linear SVM, Naive Bayes, Logistic Regression, ANN, and Long short-term memory to identify all types of attacks specifically DDoS attacks in both datasets.

Mahadik et al. [105] presented an IDS for heterogeneous IoT networks at the edge of the network by using five machine learning techniques (NB, RF, AdaBoost, ID3, LR) and one hybrid deep learning technique CNN and LSTM on the CICDDoS2019 dataset. The evaluation results show that the DL-based hybrid model gives better results than the traditional ML-based classifiers in detecting DDoS attacks having both binary class and multi-class attack classification with 100% precision, f1-score, and recall.

Nguyen et al. [106] employed a novel approach that integrates both supervised and unsupervised learning methods using a soft-ordering CNN with two outlier detectors local outlier factor and isolation-based anomaly detection using nearest-neighbor ensembles (iNNE) models. The proposed model is trained and tested based on three benchmark datasets CIC-IDS-2017, BoT-IoT, and CIC-IDS-2018 having various unknown DDoS attack scenarios and achieving a very high accuracy. The primary objective of this study is to enhance IoT network security in zero-day attack scenarios, particularly in the context of DDoS attacks. The performance of the model may diminish in real-world scenarios and also have a high false alarm rate in comparison to supervised learning models.

Deepak et al. [107] proposed a deep learning model by employing LSTM classifier for automatic feature selection and extraction and identifying DDoS attacks using the CICDDoS2019 dataset for training and testing. The accuracy obtained from the proposed framework is 98%. However, there is an absence of model deployment on real-time traffic.

Singh and Jain [108] discussed an effective Gini-Impurity-based technique for DDoS attack identification and mitigation on Software Defined Network (SDN) controllers in IoT networks. Various machine learning and deep learning classifiers Logistic Regression, Decision

Tree, K-Nearest Neighbors, and Multi-Layer Perceptron are used for detection of multi-class classification DDoS attack detection by using the dataset NSL-KDD and synthetic dataset created using the CICFlowMeter tool with detection rate between 98% to 100% and low FPR.

Aljebreen et al. [109] designed a novel approach by adopting snake optimizer and ensemble learning techniques on IoT networks using three deep learning algorithms LSTM, BiLSTM, and DBN with the Adadelta method as hyperparameter tuning. The study is trained and finally evaluated on the BoT-IoT dataset by selecting 26 features from 43 features and achieving higher performance with an accuracy of 99.8%.

Bhayo et al. [110] have designed a novel framework that is based on a Software-defined IoT ecosystem and identifies the DDoS attack detection by employing machine learning classifiers Decision Tree, Support vector machine, and Naïve Bayes on a synthetic dataset with accuracy of 98.1%, 96.1%, and 97.4% respectively. The designed framework has analyzed various parameters like payload size, IoT nodes, attack nodes, and packet frequency to find out the performance of the network through the usage of memory, detection time, and CPU usage.

12. IoT traffic datasets

Whenever any novel method is proposed by the researchers for cyber-attack detection or intrusion detection, it is implemented for its evaluation on network-based experiments. That model or method is validated with the help of various validation techniques. Four categories of such validation techniques are available which are mathematical model, simulation, emulation and real systems [111]. It has been discovered that a significant amount of current DDoS-related research in IoT networks have been using real datasets [83,87,91–93,103,110], benchmarks datasets [84,86,87,89,91–93,103,110], and simulated datasets [88,94,101] for validation of DDoS attack methods.

In the last few years, numerous initiatives have been undertaken to generate datasets having attacks targeting IoT devices. These datasets are developed to fulfill the need of researchers to generate various models for security solutions against IoT applications like healthcare, agriculture, industries, smart homes, smart power systems, etc. Different applications of IoT devices may have different needs and challenges like security, technologies, heterogeneity, etc which cannot be fulfilled by the traditional datasets. So, researchers have developed datasets

specifically by using physical or simulated IoT devices as shown in Table 5. A brief description of common datasets is as follows:

- N-Balot [112] dataset contains network data from nine IoT devices that are compromised by network intrusions. It contains 115 statistical features that could be negatively impacted by an attack.
- BoT-IoT [113] The dataset encompasses both IoT and non-IoT device traffic in the Lab of Cyber Range UNSW Canberra. After that, botnets used the traffic of compromised IoT devices. This testbed-created dataset has 46 labeled features. New features are developed to improve the performance of ML classifiers. Various dreadful attacks like DoS, DDoS, Keylogging, OS and Service Scan, and Data exfiltration exist in the BoT-IoT dataset. Furthermore, DoS and DDoS attacks are orchestrated depending upon the protocol used.
- IoTID20 [114] dataset contains network traffic using IoT devices with new flow-based features. These features help find the anomalous behavior of IoT network traffic. It includes various attacks like DOS, SCAN, ARP Spoofing, ACK flooding, SYN Flooding, SCAN Host Port, HTTP Flooding, etc.
- MedBIoT [115] dataset contains the traffic of a combination of both real and simulated IoT devices (83 devices) with 100 labeled features. The dataset can be used for IoT botnet research and Intrusion Detection for both binary-class and multi-class classification.
- IoT-23 [116] dataset was developed in January 2020 that contains network traffic using IoT devices. It has 20 instances of malware run in IoT devices and 3 captures representing benign IoT device traffic. The main target of this dataset is to provide the researcher with categorized IoT malware and benign traffic.
- TON-IoT [117] dataset comprises a variety of data sources, including network traffic datasets, Windows 7 and 10 operating system datasets, Ubuntu 14 and 18 TLS datasets, and telemetry datasets of IoT and IIoT sensors. The dataset was created at the Australian Defense Force Academy (ADFA) in the School of Engineering and Information Technology (SEIT), UNSW Canberra, at the Cyber Range and IoT Labs [118].
- Edge-IIoT [119] is a new and realistic dataset of cyber security for Internet of Things and IIoT applications. After analyzing 14 attacks against IoT and IIoT apps, the authors have suggested 61 features for the traffic. It is designed for both centralized and federated learning.

Datasets linked to the Internet of Things present unique challenges and opportunities when compared to datasets generated using non-IoT devices. Various aspects differentiate between the IoT and Non-IoT datasets like traffic behavior, features, data types, data volume, data variety, and data quality. This distinction becomes apparent through a detailed comparison of both types of datasets as shown in Table 6.

13. ML/DL software frameworks/libraries for IoT-based DDoS attack detection

There are several machine learning and deep learning tools and frameworks exist that can be used for DDoS attack detection. These tools are helpful in analyzing network traffic patterns, identifying anomalies, and enhancing the accuracy and performance of DDoS detection algorithms. Some popular machine learning and Deep learning software tools used in DDoS attack detection are described in Table 6. These tools are important for the researcher to implement and evaluate the machine learning classifiers depending on their popularity. It has been concluded from the comparative study of these tools and frameworks that TensorFlow is most widely used by researchers for training and testing purposes of deep learning classifiers as shown in 7.

13.1. MATLAB

MATLAB is a leading numerical computing platform for computational modeling and machine learning (ML), developed by MathWorks. MATLAB offers a wide range of tools and features that make it easier to create, use, and evaluate machine learning algorithms. Data preparation, feature engineering, and the application of intricate algorithms for tasks like classification, regression, clustering, and anomaly detection are just a few of the many jobs that fall within a broad category of MATLAB's machine learning capabilities. It is a potent tool in the realm of machine learning and computational modeling [120].

13.2. Weka

Weka is open-source machine learning software developed by the University of Waikato in New Zealand [121]. It is designed for data mining and predictive modeling tasks. It handles various stages of machine learning workflow like data preprocessing, classification, clustering, regression, and feature selection. Its graphical user interface is very user interactive. It has been proved to be a useful tool for machine learning exploration and application in a variety of areas, making it an invaluable resource for academic research, educational environments, and industrial applications. It also helps to compare and evaluate the results achieved from various algorithms [122].

13.3. RapidMiner

RapidMiner, a general-purpose data science software platform for text mining, ML, DL, predictive analytics, and data preparation, was developed in 2001 at the Technical University of Dortmund's Artificial Intelligence Unit [123]. It supports interactive mode (GUI), Java API, and command-line interface (CLI) [43]. RapidMiner is a widely used tool in both academia and industry that is considered the industry standard for organizations looking to use data science to make well-informed decisions [124].

13.4. Scikit-learn

Scikit-Learn is a famous Python tool which is used to design machine learning algorithms. This tool is utilized for various aspects of data mining and data analysis. It is used for development, experimentation, and deployment of machine learning models across various domains. Scikit-Learn is available in a collaboratively created Cloudera project for Hadoop clusters since April 2016 [125].

13.5. MXNet

MXNet is a deep learning framework created by the Apache Software Foundation. It is well known tool due to its efficiency, scalability and flexibility. MXNet supports a variety of neural network architectures which are suitable for distributed computing environments. It supports symbolic and imperative programming. MXNet scales well across many GPUs and various computers and is lightweight and portable. Additionally, it facilitates the effective deployment of trained models on low-end devices, including containers, serverless computing (using AWS Lambda), mobile devices [123] (using Amalgamation), and Internet of Things (using AWS Greengrass) [126].

13.6. Keras

Keras is an open-source platform that is growing quickly, with strong backend tools from major industry players like Microsoft and Google. It is a well-liked deep learning API with extensive documentation. Prominent worldwide entities like as CERN, NASA, NIH, and other international scientific institutes use Keras because of its simple and effective method for quickly developing deep learning models on a variety of backends, including TensorFlow, Theano, and CNTK [127]. Keras is excellent at encapsulating these backend libraries' functionality, offering an abstraction layer that hides their complexity [123].

Table 6

Comparisons of IoT traffic and Non-IoT traffic datasets.

Factor	Non-IoT traffic dataset	IoT traffic dataset
Traffic patterns	Traditional networks with computers and servers generates more predictable and stable traffic patterns.	Datasets with IoT devices exhibit diverse and irregular traffic patterns due to intermittent communication.
Data volume	As the number of devices is stable, data volume is comparatively very consistent. It is moderate in size with hundreds or thousands of instances.	A lot of fluctuations occur in data volume depending upon the number of devices connected. It includes thousands to billions and millions of instances in the dataset.
Device diversity	Often contains data from traditional computing devices like computers, laptops, and servers.	Contains data from a wide array of smart home appliances, industrial sensors, wearable gadgets, etc.
Payload types	Primarily involve payloads related to traditional computing tasks and applications.	May include payloads specific to IoT commands and data formats, which can differ significantly from generic data payloads.
Data quality	The dataset contains data with high quality, low noise, and few missing and incorrect data values.	The dataset has to face many data quality issues like signal interference, sensor malfunction, more noise, etc. that require more data cleaning and pre-processing methods.
Features	Include features like user identification, location, browser information, device type, etc.	The dataset includes data like humidity, temperature, motion, and information related to a particular device.

Table 7

Comparison of popular ML/DL software tools.

Tool/framework name	Release date	Developer	Language	License	Framework type	Popularity (#Stars on GitHub) As of 29-10-23
MATLAB	1984	Math Works	MATLAB	Proprietary	Machine learning	–
Weka	1993	University of Waikato	Java	GPL	Machine learning	–
Rapid Miner	2006	Rapid Miner	Java	AGPL Commercial	Machine learning	–
Scikit-learn	2007	Open-Source Community	Python	New BSD	Machine learning	56.3k
MXNet	2015	Apache Software Foundation	Python, R, Java, Perl, C++	Apache	Deep learning	36.2k
Tensor Flow	2016	Google	Python	Apache 2.0	Deep learning	178k
Keras	2015	Francois Chollet	Python	MIT	Deep learning	59.6k
PyTorch	2017	Facebook	Python	BSD	Deep learning	72.1k
Caffe	2017	BVLC	C++ with Python Interface	BSD 2-Clause	Deep learning	33.6
H2O.ai	2017	H2O.ai	Java, Python, R	Apache 2.0	Machine learning	2.6k

13.7. TensorFlow

TensorFlow is an open-source framework created by Google Brain team that utilize the data flow graphs for numerical computations for machine learning and deep learning [128]. TensorFlow may be used with single-processor systems, mobile devices, and GPUs. Various programming interfaces of TensorFlow have APIs for C++ and Python. Cloud environments from Google and Amazon also support TensorFlow. It handles multi-dimensional arrays very efficiently [123].

13.8. PyTorch

PyTorch is an open-source machine learning package for the Python programming language and designed for GPU-accelerated deep learning. It is developed by Facebook artificial-intelligence research group. It supports dynamic computational graphs which can be changed by the user during execution [129]. In PyTorch, the computation graph is defined at runtime. It has embedded and mobile framework deployment features. It supports the format Open Neural Network Exchange that is helpful to transform the models between Caffe2, PyTorch, CNTK, and other deep learning tools [123].

13.9. Caffe

Caffe is a deep learning framework developed by Berkeley Vision and Learning Center (BVLC) [130]. It has gained very popularity for its robust implementation of convolutional neural networks (CNNs). It also facilitates rapid experimentation and deployment of machine learning models. It is widely used for cyber security, image classification, segmentation, and object detection. Caffe is a valuable tool which is utilized in the dynamic field of deep learning. Data layers allow data to enter Caffe [131].

13.10. H2O

H2O is a cutting-edge open-source software for machine learning which is developed by H2O.ai. This tool is used by distributed and parallelized architecture, enabling efficient processing of extensive datasets. The software permits smooth integration into current data science workflows by supporting widely used programming languages like R and Python [132].

14. Research challenges and future directions

The present survey is a comprehensive survey on the taxonomy of IoT-based DDoS attacks, a comparative study of machine learning

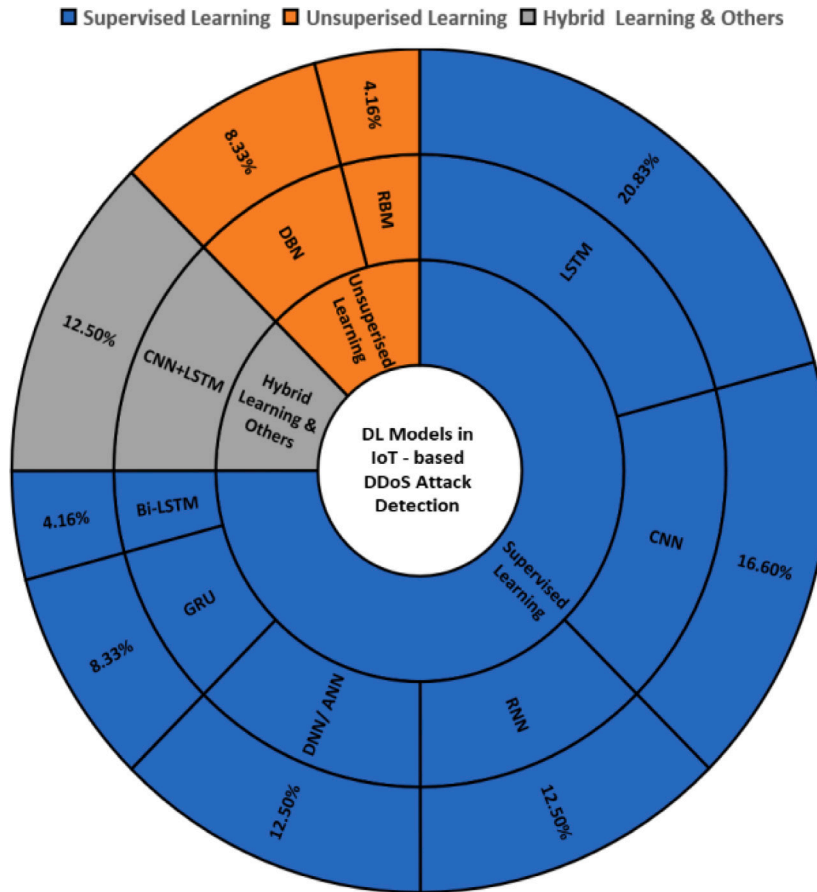


Fig. 10. Summary of DL models used in detecting IoT-based DDoS attacks.

and deep learning methodologies used for the detection of IoT-based DDoS attacks. The study encompasses datasets related to IoT which are publicly available and also gives a list of machine learning and deep learning tools and frameworks available as open source and their popularity tag. The results of our study indicate that there are some gaps and challenges which are open issues. These open issues are as follows:

- **Limited Deployment on Real-Traffic Network:** The performance of DL and ML models in prior research studies [84–89,93–95,98,102,105–108] has commonly been conducted using publicly available and benchmark datasets. These datasets use DDoS attack data for training motives that do not assure the same quality traffic data in real-time. So, implementing these models within the Real Traffic of IoT networks remains an unresolved challenge.
- **Availability of Skewed/Imbalanced Datasets:** The currently available IoT datasets mentioned in studies [112–117] exhibit an imbalance between the instances of Benign traffic and attack traffic that can adversely affect the performance of the attack detection models.
- **Limited Publicly Available IoT Datasets:** There are few publicly available datasets for attack detection in IoT networks. They might not cover the entire spectrum of IoT devices and attack scenarios. This limitation can hinder the development and evaluation of comprehensive detection models.
- **Limited work on Zero-day Attacks:** In the real world, attackers introduce novel patterns while launching new attacks that are not able to detect with the existing deployed models [133]. Therefore, it is necessary to update those models to detect unseen/Zero-Day Attacks. Also, the models should be developed that will update themselves automatically according to the new patterns of the attack behavior.

- **Limited Studies use Deep Learning:** Most of the existing studies use Machine Learning algorithms for detecting IoT-based DDoS Attacks and anomalous behavior of network traffic in IoT Networks. Since the quantity of data in an IoT environment is voluminous and complex. So, advanced algorithms, like deep learning, can effectively learn intricate patterns and detect anomalies, which will lead to more precise and reliable attack detection outcomes. The reviewed studies which have used deep learning techniques are [84–86,88,93–95,98,102,105,106].
- **Limited consideration of different Botnet families:** Most of the existing studies focus on detecting the network traffic as either Benign traffic or DDoS attack traffic. There is an absence of detection of the Malware/Botnet family under which the IoT Botnet is formed and launches the DDoS attack at the victim server.
- **Limited studies of multi-class classification of Attack Types:** Most of the existing studies are classifying the attacks in Binary Class attack categories [84–86,88–94,96,97,99,101,102,104,106–109]. A few studies [87,95,98,100,103,105,110] are classifying the DDoS attacks as either Legitimate traffic or Attack traffic as multi-class DDoS attack types (TCP flood, ICMP Flood, UDP Flood, SYN Flood, etc.). Therefore, there is a need to have a comprehensive IoT-based DDoS attack detection model for the multi-class classification of Attack traffic.
- **Older Datasets:** The literature survey includes the studies which are using different machine learning and deep learning techniques to detect DDoS attacks on older datasets like CISIDS2017, UNB-ISCX, UNSW-NB15, CICDDoS2019, NSL-KDD used in studies [85,92,95,96,102]. These older datasets have very few attack patterns and include no IoT attack diversity. Also, they possess fewer features and cannot identify different DDoS attacks effectively.

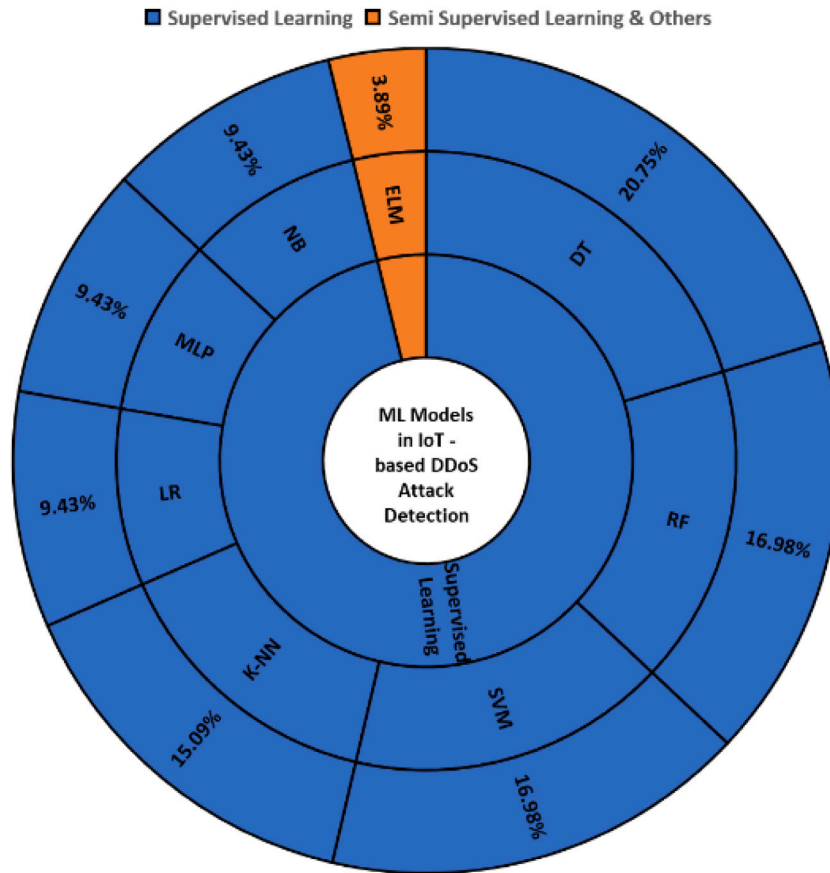


Fig. 11. Summary of ML models used in detecting IoT-based DDoS attacks.

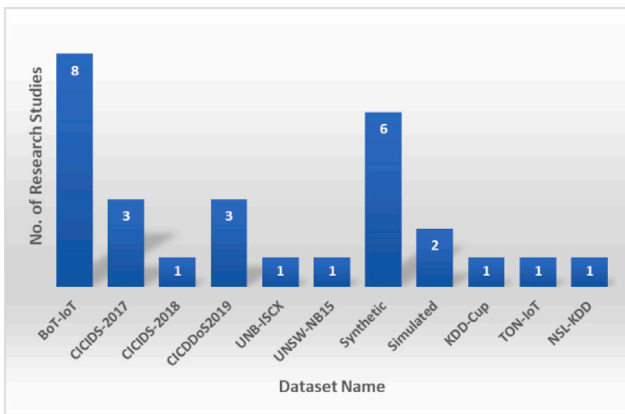


Fig. 12. Summary of datasets used by the existing research.

- **Scalability in IoT Security:** The challenges posed by the growing complexity and needs of an increasing number of IoT devices in terms of properly extending and accommodating these devices are known as scalability issues in IoT networks [134]. Some common scalability challenges in IoT networks include device proliferation, data volume and processing, network bandwidth, security concerns, Interoperability, resource constraints.

15. Conclusion and future directions

This paper illustrates IoT ecosystem and investigates various security challenges related to IoT devices at different layers of IoT architecture. The paper also discussed IoT-based DDoS attacks which is

the most destructive form of DDoS attacks in which IoT devices are compromised to amplify the scale and impact of DDoS attacks. This study draws attention to the various challenges and research issues faced by the researchers such as unavailability of the appropriate balanced real datasets with IoT traffic patterns. This study provides a comparative analysis of various machine learning and deep learning techniques proposed between 2019 to 2023 for the detection of DDoS attacks in IoT networks as AI has gained tremendous momentum in the field of security applications. The study also provides brief information regarding software frameworks and tools used to implement machine learning and deep learning classifiers. Finally, a comparative analysis of IoT traffic datasets has been done. In this paper, we have followed a systematic literature approach for the comparative analysis of machine learning and deep learning techniques to detect IoT-based DDoS attacks in the existing literature. It has been concluded that:

- (1) Deep learning and machine learning techniques are classified into various categories as shown in Fig. 9, Fig. 10 describes the percentage of studies reviewed under deep learning techniques for the detection of IoT-based DDoS attacks. In this SLR, we have reviewed 28 prominent studies from the well-known data sources. As per the proposed taxonomy of ML and DL techniques in Fig. 9, ML techniques are classified into three categories supervised learning, unsupervised learning, and semi-supervised learning. However, only few of the researchers have used unsupervised learning whereas only 4% studies have used semi-supervised learning. It has been observed that 75% of researchers have used supervised deep learning techniques, 12% of researchers have used unsupervised deep learning techniques and approximately 13% of researchers have used hybrid learning. Further, 96% of the researchers have used the supervised machine learning techniques. Decision tree (20%) classifier is extremely used by the fellow researchers (see Fig. 11).

- (2) In Section 12, various prominently used datasets in the existing literature have been summarized. It is concluded that majority of the researchers have used BoT-IoT dataset (Fig. 12) whereas CICIDS-2017 and CICDDoS2019 datasets are used by only three existing studies. Many of the real datasets such as KDD-Cup, CICIDS-2017, UNB-ISCX, etc. are obsolete nowadays. Some latest datasets like TON-IoT and Edge-IIoT are being used for testing the designed models by the researchers (see Fig. 12).
- (3) The comprehensive review concludes that 79% of selected research studies used binary DDoS attack classification that classify the traffic pattern as Malicious or Benign, 14% of the research articles used multi-class attack classification that explains various sub-categories of IoT-Based DDoS attacks, and 7% of studies cover both Binary-Class and Multi-Class attack classification.

The current findings based on the extensive literature review point to the following paths for future research:

- **AI-based Models:** Implementing an AI model for IoT security and the detection of DDoS attacks prove to be a scalable solution. These models have the potential to enhance the precision and accuracy of DDoS attack detection in IoT domain; and able to dynamically fine tune to other emerging attack patterns as well. Nevertheless, more efforts are required for the continued development and optimization of these AI models to effectively address the challenges inherent in IoT security and DDoS attack detection.
- **Balanced Datasets:** It has been observed from the systematic literature review that the availability of balanced datasets having IoT traffic patterns is limited. As a result, detection techniques exhibit bias and are unable to identify zero day attacks specifically. In the future, a more robust real IoT datasets can be generated compose of appropriate mixture of benign traffic and malicious traffic.
- **Real-time Threat Intelligence Integration:** Integrating real-time threat intelligence feeds into DDoS detection systems enables to stay updated on the latest attack signatures and trends. This dynamic approach can help identify and mitigate emerging threats effectively.
- **Need for an automated AI Model:** As the IoT-based DDoS attack patterns are changing in a great speed, there is a need for such real-time models that can be automatically updated along with the traffic patterns.

The aforementioned observations provide a foundation for the researchers to conduct studies in this domain, significantly reducing the current research challenges.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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